

Figure C-1: BG Snap Assembly Machine — Front View (IndoMIM facility)

OPERATING & SAFETY MANUAL

BG Snap Assembly Machine

IndoMIM Technologies Pvt. Ltd.

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Prepared by	Ravikumar
Designation	Certified Safety Professional
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This document is prepared in accordance with ISO 12100:2010, ISO 20607:2019, IEC 82079-1:2019, ANSI B11.19, and IS 4571.

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[IMAGE PLACEHOLDER]

Indo-MIM Limited company logo — high resolution PNG or SVG, 300 DPI minimum

OPERATING & SAFETY MANUAL

BG Snap Assembly Machine

Machine Reference: V67 — BG Snap Assembly Station

Manufacturer: Indo-MIM Limited

End User: Tata Electronics

Installation Location: Tata Electronics, MG Road, Hosur, Tamil Nadu

Document Title	Operating & Safety Manual — BG Snap Assembly Machine
Document Number	[To be assigned by Indo-MIM document control]
Revision	V1.0
Date	April 2026
Prepared By	Ravikumar, Certified Safety Professional
Status	Draft — Pending dry run verification and Open Actions closure

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1. FRONT MATTER

1.1 Document Scope

This Operating & Safety Manual covers the BG Snap Assembly Machine (Project Reference V67) installed at Tata Electronics, MG Road, Hosur, Tamil Nadu. The machine is designed and supplied by Indo-MIM Limited.

This manual covers:

- Machine identification and technical specifications
- Hazard identification and risk assessment summary
- Lockout / Tagout (LOTO) procedures for all energy isolation points
- Emergency stop function and reset procedure
- Operating instructions for all production modes including process flowchart
- Maintenance schedule — weekly, monthly, and six-monthly tasks
- Electrical, pneumatic, and hydraulic system references
- HMI alarm codes and operator responses
- Safety features, interlocks, and safety system architecture
- Compliance and certifications framework
- Residual risk summary for operator and maintenance awareness

This manual does not cover the design, manufacture, installation, or commissioning of the machine. For commissioning procedures, refer to the separate commissioning documentation held by Indo-MIM Limited.

1.2 Intended Audience

Audience	Sections to Read
All personnel working near the machine	Sections 1, 2, 3, Appendix A
Machine operators	Sections 1, 2, 3, 4, 5, 6, Appendix A
Setup technicians	All sections
Maintenance technicians	All sections — particular attention to Sections 4, 7, 12
EHS / Safety personnel	Sections 3, 4, 5, 12, 13, Appendix A
Management / Auditors	Sections 1, 3, 13, Appendix A

1.3 How to Use This Manual

Read this manual before you touch the machine. Keep it at the workstation. If a procedure is unclear, stop and ask your supervisor — do not improvise.

Safety warnings in this manual use the following signal word hierarchy as defined in ISO 20607:2019 and IEC 82079-1:2019:

DANGER

Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
[ISO 7010 — P010]

WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
[ISO 7010 — W001]

CAUTION

Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
[ISO 7010 — W017]

NOTICE

Indicates a situation that, if not avoided, may result in property or equipment damage. No personal injury risk.

1.4 General Safety Rules

The following rules apply to all personnel at all times.

1. Untrained personnel must not operate this machine. No exceptions.
2. No person may perform maintenance on this machine without completing the full LOTO procedure in Section 4 before entering the machine guarding envelope.
3. No safety device, interlock, guard, or warning label may be removed, bypassed, or modified under any circumstances without written authorization from Indo-MIM Limited and Tata Electronics EHS.
4. All incidents, near misses, and machine faults must be reported to the shift supervisor and recorded in the machine maintenance logbook immediately.
5. The machine must be operated only within the intended use defined in Section 2.3. Any use outside this scope is foreseeable misuse and is strictly prohibited.
6. Appropriate Personal Protective Equipment must be worn at all times during maintenance activities as specified in Appendix A, Section A.3.
7. This manual must be kept at or near the machine at all times and must be available to all operators and maintenance technicians during their shift.
8. When in doubt about any procedure, stop work immediately and consult the shift supervisor or maintenance technician before proceeding.

1.5 Compliance Statement Summary

This manual has been prepared in accordance with the following standards. Refer to Section 13 for the full compliance framework.

Standard	Title
ISO 12100:2010	Safety of Machinery — Risk Assessment and Risk Reduction
ISO 20607:2019	Safety of Machinery — Instruction Handbook

Standard	Title
IEC 82079-1:2019	Preparation of Instructions for Use
IS 4571:2008	Safety of Machinery — Indian Standard
Factories Act 1948	Indian Industrial Labour Law

1.6 Liability Disclaimer

This manual was prepared by Ravikumar, Certified Safety Professional, in accordance with ISO 12100:2010, ISO 20607:2019, IEC 82079-1:2019, and IS 4571:2008. It reflects the machine's design intent and known hazard profile as documented at the time of preparation. Where data was unavailable, the gaps are formally recorded as Open Actions — none have been ignored or assumed away.

Indo-MIM Limited, as machine manufacturer, and Tata Electronics, as end user, are jointly responsible for final review and verification of all information contained in this manual. Items marked as Open Actions in the working Open Actions Register must be verified and signed off before the machine is placed into production service.

Neither the manual author nor the consulting firm assumes liability for injuries or damages resulting from:

- Non-compliance with the procedures in this manual
- Operation by untrained personnel
- Unauthorized modification of the machine or its safety systems
- Failure to close Open Actions before placing the machine into service

1.7 Revision History

Revision	Date	Description	Prepared By
V1.0	April 2026	Initial issue — based on DFM V1.4 and Machine Safety Assessment Report V1.1. Placeholders retained for items pending dry run.	Ravikumar, Certified Safety Professional
V1.1	TBD	Post dry run update — LOTO verification, safety device locations, lubrication schedule, troubleshooting guide, HMI alarm codes, electrical and pneumatic schematics inserted.	Ravikumar, Certified Safety Professional
V2.0	TBD	Full release — all Open Actions closed, all placeholders replaced, customer sign-off complete.	Ravikumar, Certified Safety Professional

1.8 Table of Contents

Note: Page numbers are auto-generated by the document TOC field. Refresh the TOC in Word (right-click → Update Field → Update entire table) after any edits.

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1.9 Safety Signs Used in This Manual (ISO 7010)

This manual uses safety signs in accordance with ISO 7010:2019 — Graphical Symbols: Safety Colours and Safety Signs — Registered Safety Signs. Physical labels must be affixed to the machine by Indo-MIM Limited prior to commissioning at a minimum size of 50 × 50 mm.

Sign	Code	Description	Used in Section(s)
W001	W001	General warning	Sections 1, 3, 6
W003	W003	Electricity hazard	Sections 3, 4, 8
W017	W017	Crushing hazard	Sections 3, 5, 6
W019	W019	Automatic start-up	Sections 3, 6
W028	W028	Entanglement hazard	Sections 3, 6
P002	P002	Do not touch	Sections 3, 4
P010	P010	Do not switch on	Sections 4, 5
M008	M008	Wear protective gloves	Sections 4, 7, Appendix A
M009	M009	Wear safety footwear	Sections 4, 7, Appendix A

NOTICE

The sign graphics above are reference codes only. ISO 7010-compliant graphical labels must be sourced from an approved supplier and physically affixed to the machine at the locations specified in Section 3 and Section 12 before the machine is placed into service.

2. MACHINE OVERVIEW

2.1 Machine Identification

Field	Details
Machine Name	BG Snap Assembly Machine
Project Reference	V67 — BG Snap Assembly Station
Machine Type	Automated Snap & Screw Assembly Line
Manufacturer	Indo-MIM Limited
End User	Tata Electronics
Installation Location	Tata Electronics, MG Road, Hosur, Tamil Nadu — 631604
DFM Reference	BG Snap Assembly DFM V1.4, dated 18-March-2026
Risk Assessment Reference	Machine Safety Assessment Report V1.1, dated 23-March-2026, prepared by Independent Safety Consultant
Manual Version	V1.0
Manual Date	April 2026
Machine Weight	[PLACEHOLDER — machine weight data not available at time of manual preparation. Indo-MIM to confirm total line weight and individual station weights. See Open Action OA-P4-004.]

2.2 Technical Specifications

Parameter	Specification
Supply Voltage	220V AC (per station) / 415V, 3-phase, 50Hz (main incoming supply)
Total Connected Power	54 kW (full line)
Compressed Air Supply	0.5 – 0.7 MPa
Maximum Current	63A
Full Line Dimensions (L x W x H)	8,400 mm x 2,100 mm x 2,200 mm
Operator Headcount	1 (tray loading and unloading only)
Line Cycle Time	5 seconds per piece
Units Per Hour (UPH)	629
Design Process Yield	95% (machine in good condition, consumables within service life; material defects excluded)
Overall Equipment Effectiveness	87.42% (Availability: 93.04% / Performance: 97.03% / 1st Pass Yield: 96.84%)
Design Life	10 years (subject to scheduled maintenance and component service life)
Operating Environment	Indoor industrial use; controlled temperature and humidity; clean-room-like conditions; free from excessive dust, moisture, and corrosive atmospheres

2.3 Intended Use

This machine has one job: assemble three snap components onto an HSG unit and fasten each with two screws -- six screws per unit, every five seconds. It does nothing else. Any use outside this scope is misuse.

The machine assembles three snap components (part numbers 613-31681, 613-38625, and 613-38626) onto each HSG unit, securing each snap with two screws, for a total of six screws per assembled unit.

The machine is intended to be operated only by personnel who have been trained and authorized by Tata Electronics in accordance with the training requirements specified in Section 1.2 of this manual.

2.4 Foreseeable Misuse

WARNING

The following actions will damage the machine, produce non-conforming parts, or injure personnel. Do not perform any of the actions listed below under any circumstances.

- Operating the machine with safety guards removed or bypassed
- Manually inserting hands or tools into the machine envelope during an active cycle
- Loading HSG trays while the machine is in automatic mode without following the defined loading procedure
- Modifying robot programs, servo parameters, or safety PLC configurations without authorization from Indo-MIM Limited
- Using the machine to assemble part types or sizes other than those specified in Section 2.3
- Operating the machine in an environment outside the specified temperature, humidity, and cleanliness limits

2.5 Station Layout & Line Configuration

The full assembly line consists of six stations arranged in sequence, connected by a Beckhoff Magnetic Levitation (Maglev) conveyor system that circulates product fixtures between stations.

Station	Name	Function	Key Equipment
S1	Loading Machine	Picks HSG from input tray, scans barcode, transfers HSG into fixture on Maglev conveyor	KUKA KR13 R850 4-axis robot, barcode scanner, secondary positioning fixture
S2	Snap Assembly & Screwing Machine	Assembles first snap onto HSG and fastens with two screws; CCD vision recheck	HIWIN servo axes, Atlas QMC-21 smart screwdriver, Deprag screw feeder, tape reel feeder
S3	Snap Assembly & Screwing Machine	Assembles second snap onto HSG and fastens with two screws; CCD vision recheck	Same as S2
S4	Flipping Station	Flips HSG 180 degrees to expose reverse face for third snap assembly	Rotating cylinder, HIWIN Z-axis, pneumatic unlock mechanism

Station	Name	Function	Key Equipment
S5	Snap Assembly & Screwing Machine	Assembles third snap onto flipped HSG and fastens with two screws; CCD vision recheck	Same as S2/S3
S6	Unloading Machine	Picks completed HSG from fixture; unloads OK parts to output tray; routes NG parts to buffer or NG conveyor	KUKA KR13 R850 4-axis robot, buffer fixture, NG tray

Fixture transport: Product fixtures travel on the Maglev conveyor upper layer from S1 through S6. Empty fixtures return via the lower layer back to the loading station. RFID readers at each station link the HSG barcode to the fixture ID for full traceability throughout the assembly cycle.

2.6 Machine Layout Diagram

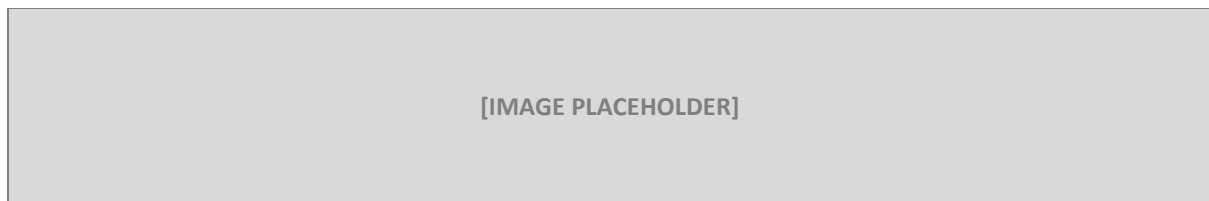


Figure 2-1: Full line top-view layout diagram showing S1 through S6 station arrangement, Maglev conveyor path, maintenance access zones, and overall machine footprint of 8,400 mm x 2,100 mm. Source: DFM V1.4 -- Machine Overview Layout. 300 DPI minimum.

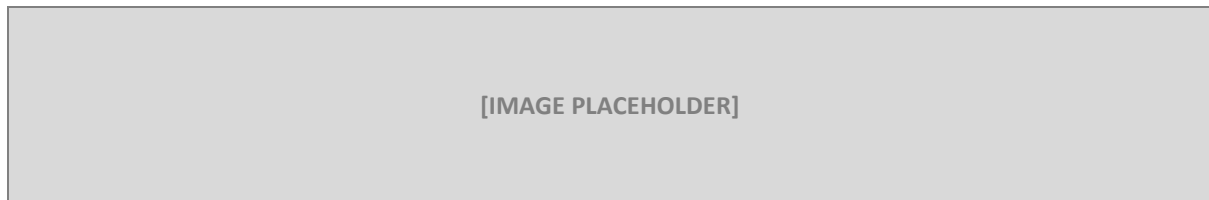


Figure 2-2: Station BOM diagram showing exploded view of all six stations with module labels S1 through S6. Source: DFM V1.4 -- Station BOM. 300 DPI minimum.

2.7 Maintenance Access Space

Keep these zones clear at all times during production. They exist for a reason -- a technician responding to a fault needs to reach the machine quickly without obstruction.

Access Zone	Clearance Required	Purpose
Front of line -- S1 and S6 ends	1,000 mm	HSG tray loading and unloading
Rear of line	1,000 mm	Snap and screw loading, general maintenance
Between S2/S3 and S5	500 mm minimum	Fixture access and channel maintenance
Top of machine	1,000 mm clearance above 2,200 mm height	Overhead service and cable management

[IMAGE PLACEHOLDER]

Figure 2-3: Maintenance space layout diagram showing all perimeter clearance zones. Source: DFM V1.4 -- Maintenance Space. 300 DPI minimum.

2.8 Key Components & Brands

Component	Brand	Model	Station
4-axis SCARA Robot	KUKA	KR 13 R850 Z200-2 CS	S1, S6
Smart Screwdriver	Atlas Copco	QMC-21-10-HM4	S2, S3, S5
Screwdriver Controller	Atlas Copco	MTF6000	S2, S3, S5
Screw Feeder	Deprag	Mini Screw Feeder 189x103x156	S2, S3, S5
Tape Reel Feeder (Snap Supply)	YIG	Tape Reel Feeder	S2, S3, S5
Maglev Conveyor	Beckhoff / Baeyer-Golytec	Magnetic Levitation Channel	Full line
Servo Axes (X/Y/Z)	HIWIN / Toyo	KK and GTH series	All stations
Safety PLC	FLX3 series	--	Full line
Safety Door Latch	Keyence	GS-51P5	S1 through S6
Safety Light Curtain	Keyence	GL-VM39FP	S1, S6
IPC & Motion Card	Advantech	IPC 7132 + PCIE-1203 series	All stations
RFID Reader	Pepperl+Fuchs	IQT1-18GM-IO-V1	All stations
Barcode Scanner	Cognex	Cognex scanner	S1
3D Snapshot Sensor	LMI	Gocator 3506	S2, S3, S5
Digital Microscope (Debug)	Dino-Lite	AM412N Portable	Full line (portable)

2.9 HMI Configuration

The machine is equipped with five operator HMI stations, one per major section of the line. Each HMI consists of a 17-inch display and a keyboard, mounted at operator-accessible height on the front face of each station enclosure.

[IMAGE PLACEHOLDER]

*Figure 2-4: HMI front-view layout diagram showing positions of five HMI displays and keyboards across S1 through S6.
Source: DFM V1.4 -- HMI Overview. 300 DPI minimum.*

HMI functions include machine start/stop, mode selection, alarm display, production data monitoring, and recipe management. Detailed HMI alarm codes and operator responses are documented in Section 11 of this manual.

OPEN ACTION — OA-P4-003

HMI alarm code list has not been provided at the time of manual preparation. Section 11 contains placeholder content. Indo-MIM to supply the complete alarm list for insertion before manual is finalized.
Responsibility: Indo-MIM Engineering | Status: OPEN

3. HAZARD IDENTIFICATION & RISK ASSESSMENT

3.1 Overview

The hazard assessment for this machine was conducted at design stage by an Independent Safety Consultant using ISO 12100:2010 methodology. Twenty hazards were identified across five categories. All have been reduced to Low residual risk. This section summarises what was found and how risk was controlled - not to satisfy paperwork, but so operators and technicians understand what they are working with.

The formal risk assessment is documented in the Machine Safety Assessment Report (V1.1, dated 23-March-2026, prepared by Independent Safety Consultant). This report is held by Indo-MIM Limited and Tata Electronics and is available for audit upon request.

This manual does not reproduce the full hazard register from the risk assessment report. The risk assessment document should be read in conjunction with this manual by safety personnel and maintenance supervisors. Residual risks remaining after all safeguards have been applied are summarised in Appendix A of this manual.

3.2 Risk Assessment Methodology

The risk assessment evaluated all foreseeable hazards associated with the BG Snap Assembly Machine across all phases of its life cycle, including normal operation, setup, changeover, troubleshooting, maintenance, and decommissioning.

Each identified hazard was scored using three factors:

Factor	Description
Severity (S)	The potential severity of injury if the hazard is realized - Minor (S1), Moderate (S2), or Serious (S3)
Exposure (E)	The frequency and duration of a person's exposure to the hazard - Prevented (E0), Low (E1), or High (E2)
Avoidance (A)	The likelihood that a person can avoid harm if exposed - Likely (A1), Not Likely (A2), or Not Possible (A3)

The combination of these three factors produces a Risk Level - Low, Medium, High, or Very High. Risk reduction measures were then applied until all residual risks reached an acceptable level.

3.3 Hazard Categories Identified

Hazard Category	Hazards Identified	Initial Risk Level	Residual Risk Level
Mechanical Hazards	10	Medium	Low (all)
Electrical Hazards	4	High (3) / Medium (1)	Low (all)
Ergonomic Hazards	2	Low	Low (all)

Hazard Category	Hazards Identified	Initial Risk Level	Residual Risk Level
Control System Hazards	4	High (3) / Low (1)	Low (all)
Chemical Hazards	0	Not applicable	Not applicable
Total	20	--	All Low

3.4 Summary of Hazard Types

3.4.1 Mechanical Hazards

Mechanical hazards arise from the movement of servo-driven axes, pneumatic cylinders, KUKA robots, the Maglev conveyor, and tray handling mechanisms.

WARNING - Tray Lifter Gravity Hazard

The tray lifter at S1 and S6 is servo-driven with no mechanical brake. When power is on, it holds position electrically. Cut the power without LOTO and it comes down. Keep hands clear unless LP-01 and LP-03 are both confirmed isolated.

Shearing hazards arise from multi-axis gantry servo systems during maintenance access. Fixed perimeter guarding and interlocked guards with Safe Torque Off (STO) on all servo drives are the primary risk reduction measures.

Entanglement and pinch hazards exist at rotating fixture assembly components, winder assemblies, and KUKA robot End-of-Arm Tooling (EOAT) during setup, troubleshooting, and maintenance. Fixed covers, interlocked safety doors, and safety-rated stops are the primary risk reduction measures.

Impact and trapping hazards exist from the KUKA SCARA robot arm and shuttle transfer mechanisms during operation and maintenance. Enclosed guarding along all robot and transfer paths, with interlocked access and STO, are the primary risk reduction measures.

Gravity-assisted crushing hazards exist from vertical lifting actuators during troubleshooting. Mechanical support or locking devices and STO with monitored brake on lifting actuators are the primary risk reduction measures.

3.4.2 Electrical Hazards

Electrical hazards arise from the presence of 415V, 3-phase, 63A supply within the machine electrical enclosures, servo drive cabinets, and robot controllers. The primary types are contact with live parts during maintenance, electric shock from insulation failure, residual stored energy after disconnection, and overheating from electrical fault conditions.

All electrical hazards are reduced to Low residual risk through a combination of enclosed electrical panel design, lockable main isolator, protective earthing, and the machine-specific LOTO procedure in Section 4 of this manual.

3.4.3 Ergonomic Hazards

Ergonomic hazards are limited to manual tray loading and unloading activities performed by the single operator. These include awkward postures arising from machine height and reach distance, and physical strain from tray weight and handling frequency. These are assessed as Low risk and are managed through operator training and correct work posture guidance in Section 6.

3.4.4 Control System Hazards

Control system hazards relate to unexpected machine start-up after power restoration, hazardous motion occurring while access doors are open, incorrect reset logic, and bypass of safety functions during maintenance. All four hazards require safety-related control functions at Performance Level PLd or PLc as specified in ISO 13849-1:2015. These are implemented through the Safety PLC and are described in Section 12 of this manual.

3.5 Risk Reduction Approach

Risk was reduced in this order - first by designing hazards out, then by guarding what could not be designed out, then by telling you what remains. This manual is the third step. The first two steps have already been done. Do not undermine them by bypassing guards or skipping LOTO.

Step 1 - Inherently Safe Design: Machine layout, enclosed station housings, and Maglev conveyor routing minimize the need for operator access to hazardous zones during normal operation. The single operator interaction point is limited to tray loading and unloading at defined access zones with physical boundary separation.

Step 2 - Safeguarding and Protective Measures: All residual hazards after inherently safe design are addressed through physical guards, interlocked safety doors (Keyence GS-51P5, PLe/Cat4/SIL3), safety light curtains (Keyence GL-VM39FP, PLe/Cat4/SIL3), Safety PLC, STO on all servo drives, dual contactors, and robot safety functions per ISO 10218-1.

Step 3 - Information for Use: Residual risks that cannot be fully eliminated by design or safeguarding are communicated through this manual, safety warning labels on the machine, operator training requirements, PPE specifications, and the maintenance procedures in Section 7.

3.6 Applicable Standards

Standard	Title	Relevance
ISO 12100:2010	Safety of Machinery - General Principles for Design	Risk assessment methodology - all sections
ISO 13849-1:2015	Safety-Related Parts of Control Systems	Safety PLC, STO, door interlocks, light curtains, E-stop circuit
ISO 13850:2015	Emergency Stop Function	E-stop design and location - Section 5
ISO 14119:2013	Interlocking Devices Associated with Guards	Safety door latch specification and interlock logic - Section 12
ISO 14120:2015	Guards - General Requirements	Fixed and interlocked guard design
ISO 14118:2017	Prevention of Unexpected Start-Up	Restart prevention logic - Section 12
ISO 13855:2024	Positioning of Safeguards	Light curtain positioning at S1 and S6

Standard	Title	Relevance
ISO 10218-1:2011	Safety Requirements for Industrial Robots	KUKA robot safety integration - Section 12
ISO 20607:2019	Instruction Handbook - Safety of Machinery	Structure and content of this manual
IEC 82079-1:2019	Preparation of Instructions for Use	Manual preparation and signal word hierarchy
IEC 60204-1:2016	Safety of Machinery - Electrical Equipment	Electrical panel design and wiring - Section 8
IEC 61496-1:2020	Electro-Sensitive Protective Equipment	Safety light curtain specification - Section 12
IS 4571:2008	Safety of Machinery (Indian Standard)	Indian regulatory compliance - all sections
IS 12100:2012	Risk Assessment - Indian Adoption of ISO 12100	Indian standards framework
Factories Act 1948	Indian Industrial Labour Law	Statutory compliance - Section 21

3.7 Residual Risk Statement

WARNING - Residual Risk

Safeguards reduce risk - they do not eliminate it. Residual risks remain on this machine. They are manageable only if LOTO is followed, PPE is worn, and the procedures in this manual are respected. Read Appendix A before your first shift.

3.8 Personnel Categories & Authorization

Personnel Category	Permitted Tasks	Training Required
Operator	Tray loading and unloading, HMI monitoring, alarm acknowledgement, normal cycle start/stop	Machine-specific operator training per this manual
Setup Technician	Recipe selection, parameter adjustment via HMI, fixture changeover	Operator training + setup and changeover training
Maintenance Technician	Scheduled and corrective maintenance, component replacement, LOTO execution	Operator training + maintenance training + LOTO certification
Safety / Authorized Person	Access to safety PLC parameters, safety function testing, RA review	All above + safety system authorization by Indo-MIM
Manufacturer's Service Engineer	Robot programming, servo parameter modification, safety PLC configuration	Manufacturer authorization only

WARNING

No personnel other than Manufacturer's Service Engineers are authorized to modify robot programs, servo drive parameters, or safety PLC configurations. Unauthorized modification of these systems may result in serious injury or death and will void the machine warranty.

4. LOCKOUT / TAGOUT (LOTO) PROCEDURES

4.1 Purpose and Scope

This section defines the Lockout / Tagout (LOTO) procedure for the BG Snap Assembly Machine. The purpose of this procedure is to ensure that the machine is completely de-energized and isolated from all hazardous energy sources before any maintenance, repair, cleaning, inspection, or fault-clearing activity is performed inside the machine guarding envelope.

This procedure applies to all personnel who are required to access the interior of the machine, including maintenance technicians, setup technicians, and authorized service engineers. Operators are not permitted to enter the machine guarding envelope under any circumstances.

This LOTO procedure has been prepared in accordance with the following standards:

- ISO 12100:2010 - Risk Assessment and Risk Reduction
- ISO 13849-1:2015 - Safety-Related Parts of Control Systems
- IEC 60204-1:2016 - Electrical Equipment of Machines
- IS 4571:2008 - Safety of Machinery (Indian Standard)
- Factories Act 1948 - Section 21 (Fencing of Machinery)

4.2 Energy Isolation Points

The BG Snap Assembly Machine has the following hazardous energy sources that must be isolated before any maintenance activity. Each isolation point is assigned a unique LOTO Point identification number.

LOTO Point ID	Energy Type	Energy Description	Isolation Device	Location on Machine
LP-01	Electrical	415V, 3-phase, 50Hz, 63A main incoming supply	Lockable rotary main isolator switch	Main electrical panel - right-hand side of S1 Loading Machine
LP-02	Pneumatic	Compressed air supply at 0.5 to 0.7 MPa	Manual quarter-turn ball valve with lockout hasp	Air inlet - rear of machine base, S1 end
LP-03	Electrical (Servo)	Servo drive Safe Torque Off (STO) - all HIWIN and Toyo axes	STO command via Safety PLC - confirmed by servo drive status indicators	Integrated in Safety PLC
LP-04	Electrical (Robot)	KUKA robot controller power - S1 and S6 robots	Robot controller main switch + teach pendant enable switch	KUKA KR CS Box-2 controller - inside S1 and S6 enclosures
LP-05	Electrical (Maglev)	Beckhoff Maglev conveyor linear motor supply	Maglev conveyor PLC isolator	Maglev control panel - centre of line

OPEN ACTION - OA-P4-001

All five LOTO isolation points listed above are identified from DFM V1.4 and the Machine Safety Assessment Report. Physical location, labelling, and hasp availability for each point must be verified during site visit at Tata Electronics, Hosur before this LOTO section is finalized. Tata Electronics maintenance team to confirm and sign off. Responsibility: Ravikumar, Certified Safety Professional - site visit | Target: Dry run | Status: OPEN

4.3 LOTO Equipment Required

Before you start, confirm you have everything below at hand. Do not begin isolation if any item is missing - get it first.

- Personal padlock (one per technician entering the machine) - keyed differently from all other padlocks
- Lockout hasp - multi-lock capable, for use at LP-01 and LP-02
- Danger tag - pre-printed with technician name, date, time, and reason for lockout
- Voltage tester - calibrated, for verifying electrical isolation at LP-01
- Pneumatic pressure gauge - for verifying air pressure has dropped to zero at LP-02
- LOTO log sheet - to record all energy isolation activities

DANGER

Your padlock. Your lock. Your life. No one else's padlock protects you. If you are inside the machine, your lock must be on every isolation point - regardless of who else has already locked out.

4.4 De-energization Sequence (Tag-Out)

The following steps must be followed in the exact order listed. Do not skip or reorder any step.

Pre-condition: The machine must be in a stopped state with all automatic cycles complete before beginning this procedure.

Step 1 - Notify

Inform all operators and personnel in the vicinity that the machine is being taken out of service for maintenance. Ensure no person is inside the machine guarding envelope before proceeding.

Step 2 - Stop the Machine

Press the HMI STOP button on the relevant station HMI to bring the machine to a controlled stop. Confirm on the HMI that all axes and robots have stopped moving and all active alarms have been acknowledged.

Step 3 - Select Maintenance Mode

At the HMI, select Maintenance Mode. This disables automatic cycle initiation and prevents remote or accidental restart from any other HMI station.

Step 4 - Isolate Main Electrical Supply (LP-01)

Turn the main rotary isolator switch at the main electrical panel to the OFF position. Apply your personal padlock through the hasp on the isolator. Attach your Danger tag to the padlock.

Step 5 - Verify Electrical Isolation (LP-01)

WARNING - Live Electrical Hazard [ISO 7010 W003]

Using a calibrated voltage tester, verify that voltage is absent at the main panel incoming terminals and at the servo drive input terminals. Do not proceed if voltage is detected.

Step 6 - Isolate Compressed Air Supply (LP-02)

Close LP-02. Apply your padlock and tag. Now wait - do not reach for a cylinder immediately. Residual air in downstream lines takes time to bleed. Watch the gauge. When it reads zero and stays there, you may proceed.

Step 7 - Verify Pneumatic Isolation (LP-02)

Confirm that the pressure gauge reads zero (0) MPa. Attempt to actuate a pneumatic cylinder manually - no movement should occur. Do not proceed if residual pressure is detected.

Step 8 - Isolate KUKA Robot Controllers (LP-04)

At each robot controller (S1 and S6), turn the controller main switch to OFF. Place the teach pendant in the docked position with the enable switch released. Apply your personal padlock to the controller cabinet door hasp.

Step 9 - Isolate Maglev Conveyor (LP-05)

At the Maglev control panel, turn the conveyor PLC isolator to OFF. Confirm that the Maglev conveyor movers are stationary and that the conveyor status indicator shows de-energized.

Step 10 - Confirm STO Status (LP-03)

Verify that the Safety PLC status indicators confirm STO is active on all servo drives. All servo drive enable indicator lights should be off.

Step 11 - Allow Stored Energy to Dissipate

WARNING - Stored Electrical Energy [ISO 7010 W003]

Wait 60 seconds. Servo drive DC bus capacitors hold charge after power is removed - enough to cause a serious shock. The 60 seconds is not a formality. Time it.

Step 12 - Record in LOTO Log

Record the following in the LOTO log sheet: technician name, date, time of isolation, list of isolation points locked out, reason for maintenance, and signature. The log sheet must remain with the machine until re-energization is complete.

NOTICE

The machine is now in a safe state. You may proceed with the authorized maintenance activity.

4.5 Re-energization Sequence (Tag-In)

The following steps must be followed in the exact order listed when returning the machine to service after maintenance is complete.

Pre-condition: All maintenance work must be fully complete. All tools, materials, and foreign objects must be removed from inside the machine guarding envelope. All guards and covers must be reinstalled and fastened.

Step 1 - Inspect and Clear

Walk the full perimeter of the machine. Verify that no tools, parts, or materials have been left inside the machine. Confirm all guards are in place and all access doors are closed. Confirm no other person is inside the machine.

Step 2 - Notify All Personnel

Inform all personnel in the vicinity that the machine is being re-energized. Ensure all persons are clear of the machine guarding envelope.

Step 3 - Restore Maglev Conveyor (LP-05)

At the Maglev control panel, turn the conveyor PLC isolator to ON. Remove your personal padlock and Danger tag from LP-05.

Step 4 - Restore KUKA Robot Controllers (LP-04)

At each robot controller (S1 and S6), turn the controller main switch to ON. Remove your personal padlock from each controller cabinet. Allow the robot controllers to complete their boot sequence before proceeding.

Step 5 - Restore Compressed Air Supply (LP-02)

Remove your personal padlock and Danger tag from LP-02. Slowly turn the pneumatic ball valve to the OPEN position. Verify air pressure builds to normal operating range (0.5 to 0.7 MPa) on the pressure gauge.

Step 6 - Restore Main Electrical Supply (LP-01)

Remove your personal padlock and Danger tag from LP-01. Turn the main rotary isolator switch to the ON position. Verify that panel indicator lights confirm power restored.

Step 7 - Confirm Safety System Status

At the Safety PLC, confirm that all safety functions are active and no safety faults are present. Confirm that all safety door interlocks are functioning correctly by opening and closing each access door and verifying the machine does not allow cycle start with any door open.

Step 8 - Reset and Return to Normal Mode

At the HMI, exit Maintenance Mode and return to Normal Operating Mode. Perform a dry run cycle without product to confirm all axes and robots move correctly before returning to production.

Step 9 - Record in LOTO Log

Record the re-energization in the LOTO log sheet: technician name, time of re-energization, confirmation that all isolation points have been restored, and signature.

NOTICE

The machine is now ready to return to service.

4.6 Energy Isolation Diagram

[IMAGE PLACEHOLDER]

Figure 4-1: LOTO Energy Isolation Diagram - full machine top-view layout showing locations of LP-01 through LP-05, colour-coded by energy type (Red = Electrical, Blue = Pneumatic). Each isolation point marked with LP-ID label and lockout hasp symbol. A3 shop-floor poster format, 300 DPI minimum. Source: To be created from DFM V1.4 layout and site visit verification. Status: Pending OA-P4-001.

4.7 Special Conditions - Partial Isolation

In certain troubleshooting situations, it may be necessary to energize specific stations while other stations remain isolated. Partial isolation is only permitted under the following conditions:

- Written authorization from the Indo-MIM maintenance supervisor and Tata Electronics safety representative
- A station-specific LOTO procedure must be documented and approved before the activity begins
- The technician performing the work must hold a valid LOTO certification
- A second authorized person must be present as safety observer at all times during partial isolation activities

WARNING - Partial Isolation Risk [ISO 7010 W001]

Partial isolation increases residual risk significantly. It must never be performed by a single person working alone, and must never be used as a routine shortcut to avoid full LOTO.

OPEN ACTION - OA-P4-006

Partial isolation procedure conditions and authorization protocol to be finalized by Indo-MIM maintenance supervisor and Tata Electronics safety representative before manual is released. Current text serves as draft framework only. Responsibility: Indo-MIM + Tata Electronics | Target: Before V2.0 | Status: OPEN

5. EMERGENCY STOP FUNCTION

5.1 Purpose and Scope

The E-stop is a last resort. It is not a pause button, not a production stop, and not a substitute for LOTO. Use it when something unexpected is happening and you need everything to stop now. For planned maintenance, use LOTO.

The emergency stop function has been designed in accordance with **ISO 13850:2015 - Safety of Machinery: Emergency Stop Function - Principles for Design**.

5.2 E-Stop Device Locations

The BG Snap Assembly Machine is equipped with emergency stop pushbuttons at the following locations. All E-stop devices are red mushroom-head pushbuttons on a yellow background, conforming to ISO 13850:2015 and IEC 60947-5-5.

E-Stop ID	Location	Station	Operator Access
ES-01	HMI control panel - left side	S1 Loading Machine	Front of machine
ES-02	HMI control panel - left side	S2 Snap Assy & Screwing	Front of machine
ES-03	HMI control panel - left side	S3 Snap Assy & Screwing	Front of machine
ES-04	Flipping station control panel	S4 Flipping Module	Front of machine
ES-05	HMI control panel - left side	S5 Snap Assy & Screwing	Front of machine
ES-06	HMI control panel - left side	S6 Unloading Machine	Front of machine

OPEN ACTION - OA-P4-007

Physical E-stop locations, exact mounting heights, and panel positions to be verified during site visit at Tata Electronics, Hosur. Locations listed above are derived from DFM V1.4 HMI layout. Site verification required before this section is finalized. Responsibility: Ravikumar, Certified Safety Professional - site visit | Target: Dry run | Status: OPEN

[IMAGE PLACEHOLDER]

Figure 5-1: E-stop location diagram - front view of full machine line showing ES-01 through ES-06 positions marked with red indicators. Source: To be created from DFM V1.4 HMI layout and site visit photographs. 300 DPI minimum.

5.3 Stop Category

The emergency stop function of the BG Snap Assembly Machine is implemented as a **Category 1 Stop** as defined in **IEC 60204-1:2016 - Safety of Machinery: Electrical Equipment of Machines**.

Stop Category	Definition	Implementation on this Machine
Category 1	Controlled stop - power is maintained to machine actuators to achieve a controlled deceleration to standstill, after which power is removed	All servo drives decelerate to standstill under controlled braking, then Safe Torque Off (STO) is applied. KUKA robots execute a controlled stop before power removal. Pneumatic supply remains energized during deceleration to maintain cylinder positions, then dumps to safe state.

This stop category is selected because the BG Snap Assembly Machine operates high-speed servo axes and KUKA robots that require controlled deceleration to avoid mechanical damage, product damage, and secondary hazards from uncontrolled stopping.

5.4 E-Stop Function Description

When any E-stop pushbutton is activated, the following sequence occurs automatically:

Step 1 - Immediate Signal to Safety PLC

The E-stop signal is transmitted to the Safety PLC via a dual-channel hardwired safety circuit. The Safety PLC processes the signal and initiates the controlled stop sequence within the safety response time of the system.

Step 2 - Controlled Deceleration of All Axes

All HIWIN and Toyo servo axes decelerate to standstill under controlled braking. All KUKA robots execute a Category 1 controlled stop per ISO 10218-1.

Step 3 - Safe Torque Off (STO) Applied

Once all axes reach standstill, the Safety PLC applies STO to all servo drives simultaneously. Drive enable signals are removed. The Maglev conveyor is de-energized.

Step 4 - Pneumatic Safe State

The pneumatic system dumps to its fail-safe state. Cylinders holding product or fixtures retain their position through mechanical means where designed; all other pneumatic actuators move to their de-energized position.

Step 5 - HMI Alarm Display

All five HMI stations display an E-stop active alarm with the station identifier of the activated E-stop device. The alarm remains active until the E-stop is reset and the alarm is acknowledged.

5.5 E-Stop Reset Procedure

The emergency stop function is a latching function. Once activated, it cannot be reset remotely or automatically. The following steps must be completed by an authorized operator or technician before the machine can return to normal operation.

WARNING - Before Resetting E-Stop [ISO 7010 W019]

The machine will move immediately on restart. Before you press RESET, walk the machine. Find out why the E-stop activated. If you cannot identify the cause, do not reset - call the maintenance technician.

Step 1 - Identify Cause

Walk the full perimeter of the machine and identify the reason for the E-stop activation. Resolve any hazardous condition, remove any obstruction, and confirm that all personnel are clear of the machine guarding envelope.

Step 2 - Release the E-Stop Pushbutton

At the station where the E-stop was activated, rotate the mushroom-head pushbutton clockwise until it releases and returns to its extended position. The E-stop pushbutton must be physically released at the device - it cannot be released from the HMI.

Step 3 - Acknowledge Alarm at HMI

At the HMI of the station where the E-stop was activated, acknowledge the E-stop alarm. Confirm that no other active alarms are present before proceeding.

Step 4 - Confirm Safety System Status

Confirm on the HMI that the Safety PLC reports all safety functions as active and healthy. Confirm that all safety door interlocks are engaged and all light curtains are unobstructed.

Step 5 - Perform Restart Authorization

Only an authorized operator or technician may authorize machine restart after an E-stop event. The restart must be initiated from the HMI using the defined restart sequence - the machine will not restart from an automatic cycle command alone.

Step 6 - Initiate Controlled Restart

Press the RESET button on the HMI followed by the CYCLE START button. The machine will perform a controlled homing sequence before resuming normal operation. Monitor the first cycle carefully for any abnormal motion or alarm conditions.

Step 7 - Record the E-Stop Event

Record the E-stop event in the machine maintenance log, including the station, time, cause, corrective action taken, and name of the person who performed the reset. This record is required for maintenance trend analysis and regulatory compliance.

5.6 Periodic Testing Requirements

The E-stop function must be tested at the following intervals to verify correct operation. Test results must be recorded in the machine maintenance logbook with the date, technician name, result, and acceptance criteria used.

Test	Frequency	Procedure	Acceptance Criteria
Functional test - all E-stop buttons	Weekly	Activate each E-stop pushbutton individually during a test cycle. Verify machine stops within expected response time.	All axes stop, STO applied, HMI alarm displayed, machine does not restart without manual reset
Dual-channel integrity test	Monthly	Disconnect one channel of the E-stop circuit at the Safety PLC input terminal and verify the Safety PLC detects the fault and generates an alarm	Safety PLC fault alarm generated within 500ms
Response time verification	Every 6 months	Use a calibrated stopwatch or data logger to measure time from E-stop activation to full axis standstill	Standstill achieved within manufacturer-specified deceleration time
Full safety function audit	Annually	Complete safety function test per ISO 13849-1 verification checklist, conducted by authorized safety technician	All safety functions verified as meeting PLd/Category 3 requirements

5.7 What the E-Stop Does NOT Do

Know what pressing E-stop does not do. It does not cut 415V from the electrical panel. It does not fully vent pneumatic pressure. It does not lock out the machine. It does not stop someone else from restarting the machine. For any of those outcomes, use LOTO.

Specifically:

- The E-stop does not isolate the main electrical supply (415V). The main electrical panel remains energized after an E-stop activation. For electrical isolation, the full LOTO procedure in Section 4 must be followed.
- The E-stop does not vent all pneumatic pressure from the system immediately. Residual pressure may remain in pneumatic lines. Verify pressure at zero before accessing pneumatic components.
- The E-stop does not substitute for the LOTO procedure. Never use the E-stop as a lockout device for maintenance activities.
- The E-stop does not prevent re-energization by another person. Always apply your personal LOTO padlock when entering the machine guarding envelope.

6. OPERATING INSTRUCTIONS

6.1 Purpose and Scope

During automatic operation your role is simple: load trays at S1, collect trays at S6, monitor the HMI. The machine handles everything else. Do not interfere with the automatic cycle. If something looks wrong, press E-stop and call the technician.

This section covers all operator activities during normal production, including machine startup, tray loading and unloading, cycle monitoring, and machine shutdown.

Operators must have completed the machine-specific training programme approved by Tata Electronics before being permitted to operate this machine independently.

6.2 Operating Modes

The BG Snap Assembly Machine has three operating modes selectable from the HMI.

Mode	Description	Who Can Select
Automatic Mode	Full automatic assembly cycle. Machine runs continuously without operator intervention except for tray loading and unloading.	Trained Operator
Manual Mode	Individual station or axis movements can be commanded one step at a time from the HMI. Used for setup, changeover, and troubleshooting only.	Setup Technician or above
Maintenance Mode	All automatic functions disabled. Used during maintenance activities in conjunction with the LOTO procedure in Section 4.	Maintenance Technician or above

WARNING - Manual Mode Restriction [ISO 7010 W019]

Manual Mode allows individual machine movements with some guards in a monitored state. Manual Mode must never be used for production operation. Only authorized Setup Technicians and above may select Manual Mode.

6.3 Pre-Start Checklist

Run through this list at the start of every shift. It takes three minutes. Skipping it takes longer to explain after something goes wrong.

#	Check Item	Required Condition
1	All access doors and safety guards	Fully closed and latched
2	All E-stop pushbuttons (ES-01 to ES-06)	Released - not activated (buttons in extended position)
3	Safety light curtains at S1 and S6	Unobstructed - green indicator active

#	Check Item	Required Condition
4	Compressed air pressure gauge at inlet	Reading between 0.5 and 0.7 MPa
5	Main electrical isolator (LP-01)	In ON position
6	All HMI displays	Powered on - no active alarms
7	KUKA robot controllers (S1 and S6)	Powered on - no fault indicators
8	Maglev conveyor	Powered on - conveyor status normal on HMI
9	Input tray area at S1	Clear of obstructions - tray loaded with HSG units
10	Output tray area at S6	Empty tray in position - ready to receive completed assemblies
11	Screw feeders (S2, S3, S5)	Loaded with screws - feeder level indicator showing adequate supply
12	Tape reel feeders (S2, S3, S5)	Snap tape reels loaded and threaded correctly
13	LOTO log	No open LOTO entries - machine released from maintenance
14	Work area around machine	Clear of personnel, tools, and obstructions

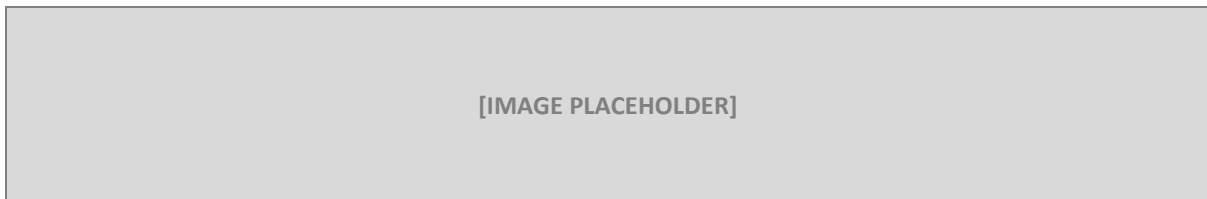


Figure 6-1: Pre-start checklist reference photograph - front view of machine showing key check points numbered 1 through 14 with callout arrows. Source: To be captured during site visit at Tata Electronics, Hosur. 300 DPI minimum.

6.4 Machine Startup Procedure

Pre-condition: All pre-start checks in Section 6.3 must be completed and confirmed before beginning startup.

Step 1 - Power On

Confirm the main rotary isolator at LP-01 is in the ON position. At each HMI station, confirm the display is active and showing the home screen. If any HMI is blank, inform the maintenance technician before proceeding.

Step 2 - Clear All Alarms

At each HMI station (S1 through S6), check the alarm screen. Acknowledge and clear any residual alarms from the previous shift. If any alarm cannot be cleared, do not proceed - contact the maintenance technician.

Step 3 - Initialize Safety System

At the S1 HMI, navigate to the Safety System status screen. Confirm that the Safety PLC reports all safety functions as healthy. Confirm that all safety door interlocks show as engaged and both light curtains show as active.

Step 4 - Home All Axes

At the S1 HMI, select the Homing function and initiate the automatic homing sequence. The machine will move all servo axes to their home positions in a defined sequence. Monitor the homing sequence on the HMI. Do not interrupt the homing sequence once started.

Step 5 - Initialize KUKA Robots

Confirm on the HMI that both KUKA robots (S1 and S6) have completed their initialization sequence and are reporting ready status. If either robot shows a fault, contact the maintenance technician before proceeding.

Step 6 - Initialize Maglev Conveyor

At the Maglev control panel, confirm the conveyor is initialized and all fixture movers are detected. The HMI should show the correct number of movers in circulation.

Step 7 - Select Recipe

At the S1 HMI, navigate to the Recipe screen. Select the correct product recipe for the current production run. Confirm the recipe name and version number match the production order. An incorrect recipe will result in wrong assembly parameters and non-conforming product.

Step 8 - Select Automatic Mode

At the S1 HMI, select Automatic Mode. Confirm that the mode indicator on all HMI stations changes to Automatic.

Step 9 - Load Input Tray

Place a fully loaded input tray of HSG units into the tray storage module at S1. Ensure the tray is seated correctly in the tray guide rails. Push the tray fully into position until the tray clamping jaws engage audibly. The HMI should display tray detected status.

Step 10 - Start Cycle

Press the CYCLE START button on the S1 HMI. The machine will begin the automatic assembly cycle. Monitor the first cycle on the HMI to confirm all stations are operating normally before leaving the machine unattended.

6.5 Normal Operating Cycle - Description

During normal automatic operation the machine executes the following sequence continuously without operator intervention. The operator's only responsibilities during automatic operation are tray loading at S1 and tray unloading at S6.

S1 - Loading:

The KUKA robot picks one HSG unit from the input tray, places it into the secondary positioning fixture for barcode scanning and pre-alignment, then transfers it into a Maglev fixture mover. If the barcode scan fails, the HSG is automatically routed to the NG conveyor without entering the assembly sequence.

S2 - First Snap Assembly & Screwing:

The fixture mover carries the HSG into S2. The tape reel feeder advances one snap component. The voice coil actuator picks the snap, the CCD camera performs a pre-inspection check, and the snap is placed onto the HSG at the first assembly position. The Atlas smart screwdriver picks two screws from the Deprag screw feeder and drives them to the specified torque. The lower CCD camera rechecks screw seating after driving. If torque or seating is NG, the screws are tossed and the HSG is flagged.

S3 - Second Snap Assembly & Screwing:

The fixture mover advances to S3. The same snap and screw assembly sequence as S2 is repeated for the second snap position on the HSG.

S4 - Flipping:

The fixture mover advances to S4. The flipping module suctions the HSG from the fixture, rotates it 180 degrees, and replaces it back into the fixture with the reverse face exposed for third snap assembly.

S5 - Third Snap Assembly & Screwing:

The fixture mover advances to S5. The same snap and screw assembly sequence as S2 and S3 is repeated for the third snap position on the reverse face of the HSG.

S6 - Unloading:

The KUKA robot picks the completed HSG from the fixture. If the assembly result is OK across all three stations, the HSG is placed into the output tray. If the assembly result is NG at any station, the HSG is routed to the buffer fixture or NG tray for segregation. The operator collects completed output trays and replaces them with empty trays.

6.6 Normal Operating Cycle - Process Flowchart

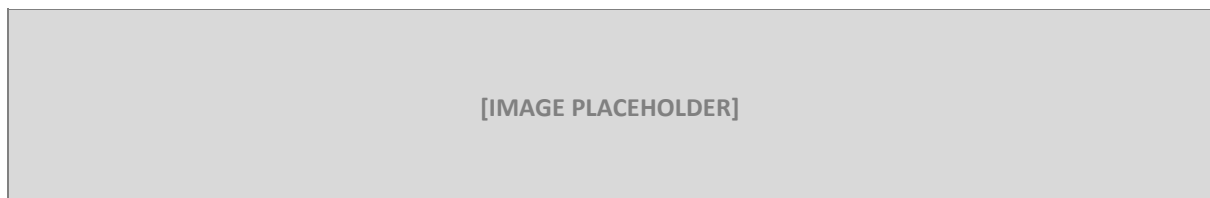


Figure 6-2: Normal operating cycle process flowchart - full assembly sequence from HSG loading at S1 through S2, S3, S4 (flip), S5, to unloading at S6, with OK and NG decision paths at barcode scan, torque and seating check, and final all-stations check. Colour scheme: dark blue #1F3864 process boxes, #D9E1F2 fill, white decision diamonds with dark blue border, dashed NG rejection paths. Legend included lower left. Full page width, 300 DPI minimum.

6.7 Tray Loading & Unloading Procedure

This is the operator's primary interaction with the machine during automatic operation. All other machine functions are fully automatic.

Tray Loading at S1 (Input):

Step 1 - When the HMI at S1 displays a Tray Empty or Load New Tray prompt, approach the S1 loading zone. Confirm the safety light curtain indicator at S1 is green before approaching.

Step 2 - Pick up a fully loaded input tray of HSG units from the supply rack. Verify the tray is correctly loaded - HSG units seated flat, no tilted or missing units.

Step 3 - Insert the tray into the tray storage module at S1, aligning it with the tray guide rails. Push the tray fully into position until the tray clamping jaws engage audibly.

Step 4 - Confirm on the HMI that the tray detected status is displayed. Do not reach into the machine beyond the tray loading slot.

Step 5 - Step back behind the safety light curtain line. The machine will automatically begin picking HSG units from the new tray.

WARNING - KUKA Robot Reach Zone at S1 [ISO 7010 W017]

The KUKA robot at S1 reaches the tray loading zone at full speed with no warning. The light curtain will stop the machine if you break the beam - but a curtain trip is not the same as being safe. Stay behind the defined loading line. Load the tray, step back. That is the complete interaction.

Tray Unloading at S6 (Output):

Step 1 - When the HMI at S6 displays an Output Tray Full or Remove Tray prompt, approach the S6 unloading zone.

Step 2 - Remove the completed output tray from the output tray storage module. Place it on the output rack for transfer to the next process.

Step 3 - Insert an empty output tray into the S6 storage module and confirm the tray is seated correctly.

Step 4 - Confirm on the HMI that the empty tray is detected. Step back behind the safety light curtain line.

Step 5 - Check the NG buffer tray periodically. Remove and segregate any NG parts in the buffer fixture or NG tray. Label NG parts clearly and follow Indo-MIM's non-conforming product procedure.

6.8 Normal Shutdown Procedure

Use this procedure at the end of every shift or when stopping the machine for a planned break.

Step 1 - Let the current cycle finish before stopping. Mid-cycle stops leave HSG units partially assembled inside fixtures - these become NG parts and require manual clearance by a technician before the next run.

Step 2 - At the S1 HMI, press the CYCLE STOP button. The machine will complete the current cycle and then come to a controlled stop.

Step 3 - Remove any partially loaded input trays from S1. Remove any completed output trays from S6. Remove and segregate any NG parts from the buffer tray.

Step 4 - At the HMI, navigate to the shutdown sequence and select Controlled Shutdown. The machine will home all axes, de-energize the Maglev conveyor, and power down the robot controllers in a defined sequence.

Step 5 - Once the HMI confirms shutdown complete, turn the main rotary isolator at LP-01 to the OFF position if required by site procedure.

Step 6 - Record any alarms, faults, or observations from the shift in the machine logbook.

6.9 Emergency Shutdown

Use this procedure only when an immediate stop is required due to an unsafe condition.

Step 1 - Press the nearest E-stop pushbutton (ES-01 through ES-06). The machine will execute a Category 1 controlled stop - all axes decelerate and STO is applied within the safety response time.

Step 2 - Do not attempt to restart the machine until the cause of the emergency stop has been identified and resolved.

Step 3 - Follow the E-stop reset procedure in Section 5.5 before returning the machine to service.

Step 4 - Record the emergency stop event - time, station, cause, and corrective action - in the machine logbook. Report the event to the shift supervisor immediately.

CAUTION - After Any E-Stop [ISO 7010 W001]

After any E-stop, assume something has moved to an unexpected position. A fixture that stopped mid-travel, a screw that was not fully driven, a snap that was half-placed - any of these will cause a fault or jam on restart. Look before you reset.

7. MAINTENANCE & CARE

7.1 Purpose and Scope

This section defines the maintenance requirements for the BG Snap Assembly Machine. It covers scheduled preventive maintenance, lubrication requirements, consumable replacement, cleaning procedures, and basic troubleshooting guidance for maintenance technicians.

All maintenance activities must be performed by trained and authorized maintenance technicians only. Before entering the machine guarding envelope for any maintenance task, the full LOTO procedure defined in Section 4 of this manual must be completed without exception.

This maintenance schedule has been prepared based on the machine design documentation (DFM V1.4) and industry best practices for servo-driven automated assembly equipment. Maintenance intervals should be reviewed and updated by Indo-MIM's engineering team after the first year of operation based on actual machine performance data.

DANGER - LOTO Required for All Machine Access [ISO 7010 P010]

Every task in this section that requires entering the machine requires LOTO first. This is not repeated for every task - it is stated once, here, and it applies everywhere. No exceptions for quick jobs. No exceptions for familiar tasks. No exceptions.

7.2 Maintenance Personnel Requirements

Activity	Minimum Qualification
Visual inspection and cleaning	Trained operator under supervision of maintenance technician
Consumable replacement (screws, snap reels, nozzles)	Trained maintenance technician
Lubrication of linear axes	Trained maintenance technician
Fixture inspection and replacement	Trained maintenance technician
Robot calibration check	Authorized service engineer (Indo-MIM or KUKA)
Safety function testing	Authorized safety technician
Safety PLC parameter changes	Manufacturer's authorized engineer only
Vision system calibration	Trained maintenance technician with vision system training

7.3 Scheduled Maintenance - Overview

Three maintenance frequencies. Weekly keeps the machine clean and the sensors reliable. Monthly keeps the drives accurate and the safety functions verified. Six-monthly keeps the machine compliant and the major wear items replaced before they fail. Miss any level and the levels above it become meaningless.

The pre-shift checks required before every production shift are covered separately in Section 6.3 - Pre-Start Checklist of this manual.

7.4 Weekly Maintenance Tasks

Perform at the start of the first shift of each working week. Machine must be in LOTO state for tasks marked LOTO REQUIRED.

Task #	Task Description	Station	LOTO Required	Method
W-01	Test all E-stop pushbuttons (ES-01 to ES-06) individually - verify controlled stop and HMI alarm display	All	No	Activate each button during a test cycle; verify stop and alarm
W-02	Inspect all fixed guards and enclosure panels - verify all fasteners are tight	All	No	Visual and tactile check; tighten any loose fasteners
W-03	Clean CCD camera lenses at S2, S3, S5 - upper and lower cameras	S2,S3,S5	LOTO REQUIRED	Lint-free cloth and approved optical cleaning solution only. A single fingerprint on a telecentric lens will cause snap pre-check failures that look like a snap feeding fault. Clean the lens before assuming the feeder is the problem.
W-04	Clean LMI Gocator 3D snapshot sensor windows at S2, S3, S5	S2,S3,S5	LOTO REQUIRED	Lint-free cloth only - do not use solvents
W-05	Inspect HIWIN and Toyo linear axis rails - remove debris and check for contamination	All	LOTO REQUIRED	Visual check and wipe with clean dry cloth
W-06	Inspect KUKA robot EOAT grippers and nozzles - check for wear, damage, or contamination	S1, S6	LOTO REQUIRED	Visual inspection; replace if worn or cracked
W-07	Inspect snap tape reel feeder mechanism - clean feed path, check for tape debris	S2,S3,S5	LOTO REQUIRED	Visual inspection and clean with dry compressed air
W-08	Inspect Deprag screw feeder - clean feed track, check for jammed or deformed screws	S2,S3,S5	LOTO REQUIRED	Visual inspection; remove jammed screws; clean track
W-09	Inspect Atlas smart screwdriver nozzles - check for wear and screw seating integrity	S2,S3,S5	LOTO REQUIRED	Visual inspection; replace nozzle if tip is worn or damaged
W-10	Inspect Maglev conveyor mover carriages - check for debris on rail surface	Full line	LOTO REQUIRED	Visual check; clean rail surface with dry lint-free cloth

Task #	Task Description	Station	LOTO Required	Method
W-11	Inspect fixture bodies - check push blocks, datum blocks, and clamping surfaces for wear	All	LOTO REQUIRED	Visual and dimensional check; replace worn components
W-12	Check pneumatic tubing and fittings for leaks or damage	All	No	Listen for air leaks during pressurized operation; visual check
W-13	Verify RFID reader status at all six stations - confirm correct fixture ID reading	All	No	Run test cycle; confirm RFID read confirmation on HMI

7.5 Monthly Maintenance Tasks

Perform at the start of the first week of each calendar month. Full LOTO required for all tasks unless stated.

Task #	Task Description	Station	Method
M-01	Lubricate HIWIN linear guide rail blocks - apply recommended grease via grease nipple	All	Per lubrication schedule - see Section 7.8
M-02	Lubricate Toyo servo axis ball screw - apply recommended grease	All	Per lubrication schedule - see Section 7.8
M-03	Inspect and clean Pepperl+Fuchs RFID reader faces - check mounting and cable condition	All	Wipe with dry cloth; confirm secure mounting
M-04	Inspect Keyence safety door latch mechanisms - check latch pin condition and actuator alignment	All	Manual operation check; replace if latch pin shows wear
M-05	Test dual-channel integrity of E-stop circuit at Safety PLC input terminals	All	Disconnect one channel; verify Safety PLC fault alarm within 500ms
M-06	Check servo drive cooling fan operation - confirm airflow from drive cabinet ventilation	All	Visual and tactile; listen for abnormal fan noise
M-07	Inspect pneumatic cylinder seals at tray locking and flipping modules - check for air weeping	S1,S4,S6	Listen and visual; replace cylinder if seal weeping detected
M-08	Check Maglev conveyor mover wheel condition - inspect for flat spots or uneven wear	Full line	Visual check with movers stationary
M-09	Verify Atlas screwdriver torque calibration - run calibration check per Atlas MTF6000 procedure	S2,S3,S5	Atlas calibration tool; confirm torque within plus or minus 5% of set value. Torque drift is the most common silent defect on this machine - the screwdriver drives, the CCD passes the screw position, but the torque is out of spec. The

Task #	Task Description	Station	Method
			customer sees it; you do not. Calibrate monthly without fail.
M-10	Inspect IPC and motion card cooling - clean dust filters on Advantech IPC cabinets	All	Remove filter; clean with compressed air; reinstall
M-11	Back up HMI recipe files and PLC program to approved storage location	All	HMI backup function; store to designated USB or network location
M-12	Inspect voice coil actuator stroke - verify correct snap pick-up and placement position	S2,S3,S5	Test run in Manual Mode; confirm pick and place positions

7.6 Six-Monthly Maintenance Tasks

Perform every six months. Full LOTO required. Some tasks require authorized specialist personnel.

Task #	Task Description	Station	Specialist Required
SM-01	Full safety function audit - verify all PLe/Cat4 safety functions per ISO 13849-1 verification checklist	All	Authorized safety technician
SM-02	KUKA robot full calibration check - verify TCP, axis positions, and working envelope	S1, S6	KUKA authorized service engineer
SM-03	Vision system calibration - full calibration of all CCD cameras using calibration plate	S2,S3,S5	Vision-trained maintenance technician
SM-04	LMI Gocator 3D sensor calibration - full calibration using reference target	S2,S3,S5	Vision-trained maintenance technician
SM-05	Replace Atlas screwdriver nozzles - scheduled replacement regardless of visual condition	S2,S3,S5	Maintenance technician
SM-06	Replace fixture push blocks (PU material) - scheduled replacement due to wear	All	Maintenance technician
SM-07	Full electrical inspection - check all cable terminations, earth continuity, and insulation resistance	All	Qualified electrician
SM-08	Maglev conveyor full system check - verify mover positioning accuracy and rail alignment	Full line	Beckhoff authorized technician
SM-09	E-stop response time verification - measure time from E-stop activation to full axis standstill	All	Authorized safety technician
SM-10	Review and update Open Actions Register - close verified items; add new observations	All	Certified Safety Professional or Third Party

7.7 Maintenance Records

All maintenance activities must be recorded in the machine maintenance logbook kept at the machine. Each entry must include:

- Date and time of maintenance activity
- Task number and description referencing the schedule in Sections 7.4 to 7.6
- Name and signature of technician performing the work
- Result of the activity - satisfactory, item replaced, fault found, or deferred
- Any observations, measurements, or unusual findings noted during the task
- For safety function tests - the measured result and the acceptance criteria used

NOTICE - Legal Retention Requirement

Maintenance records are a legal document under the Factories Act 1948 and must be retained by Tata Electronics for a minimum of three years. Records must be available for inspection by a Certified Safety Professional or Third Party, Indo-MIM quality team, or statutory authorities upon request.

7.8 Lubrication Schedule

OPEN ACTION - OA-P4-008

The lubrication schedule for this machine - including lubricant types, approved brands, application quantities, and confirmed service intervals for all HIWIN axes, Toyo axes, KUKA robot joints, SMC pneumatic cylinders, and rotating cylinder pivot points - is pending confirmation with Indo-MIM engineering team. Responsibility: Indo-MIM Engineering | Target: Before manual finalization | Status: OPEN

[IMAGE PLACEHOLDER]

Figure 7-1: [PLACEHOLDER] Full lubrication schedule table - Component | Lubricant Type | Approved Brand | Application Point | Interval | Quantity per application. To be inserted following confirmation with Indo-MIM.

[IMAGE PLACEHOLDER]

Figure 7-2: Lubrication point diagram - annotated layout showing all lubrication points across all stations with LP identification labels. Source: To be created from DFM V1.4 module detail drawings following Indo-MIM confirmation. 300 DPI minimum.

CAUTION - Lubrication Quantity

Do not over-grease linear guide blocks. Excess grease attracts metal particles and contaminates the guide rail, accelerating wear. Apply only the quantity specified in the confirmed lubrication schedule.

7.9 Consumables & Spare Parts

Item	Part Description	Station	Replacement Trigger	Minimum Stock
Snap components	BG Snap - part nos. 613-31681, 613-38625, 613-38626	S2,S3,S5	Tape reel empty	As per production plan
Screws	M-type screws per assembly specification	S2,S3,S5	Feeder at minimum level	2 shift supply minimum
Atlas screwdriver nozzle tips	QMC-21-10-HM4 compatible nozzles	S2,S3,S5	Worn or damaged / 6-monthly schedule	4 sets minimum
Fixture push blocks (PU)	Long side push block, short side push block	All	Worn or deformed / 6-monthly schedule	2 sets per station
Fixture datum blocks (PPS)	Long side datum, short side datum	All	Cracked or dimensionally out of tolerance	1 set per station
Tape reel feeder drive belt	YIG tape reel feeder compatible belt	S2,S3,S5	Worn or slipping	2 per feeder type
CCD camera cleaning cloths	Lint-free optical cleaning cloth	S2,S3,S5	Weekly use	1 pack minimum
Safety door latch	Keyence GS-51P5 complete latch assembly	All	Latch pin worn / interlock fault	2 units minimum

7.10 Basic Troubleshooting Guide

OPEN ACTION - OA-P4-009

The troubleshooting guide for this machine - covering common fault conditions, probable causes, and step-by-step corrective actions - is pending confirmation with Indo-MIM engineering team and Tata Electronics maintenance team following the site visit at Tata Electronics, Hosur. Responsibility: Indo-MIM Engineering + Tata Electronics Maintenance | Target: Before manual finalization | Status: OPEN

[IMAGE PLACEHOLDER]

Figure 7-3: [PLACEHOLDER] Full troubleshooting table - Fault Description | HMI Alarm Code | Probable Cause | Corrective Action. To be populated from site visit observations, operator interviews, and Indo-MIM engineering input.

8. ELECTRICAL SCHEMATIC

8.1 Purpose

This section provides the electrical schematic documentation for the BG Snap Assembly Machine, including the main power distribution circuit, control circuit, safety circuit, and cable schedule.

8.2 Electrical Schematic Drawings

OPEN ACTION - OA-P4-002

Electrical schematic drawings for the BG Snap Assembly Machine have not been provided at the time of preparation of this manual. Responsibility: Indo-MIM Engineering | Target: Before manual finalization | Status: OPEN

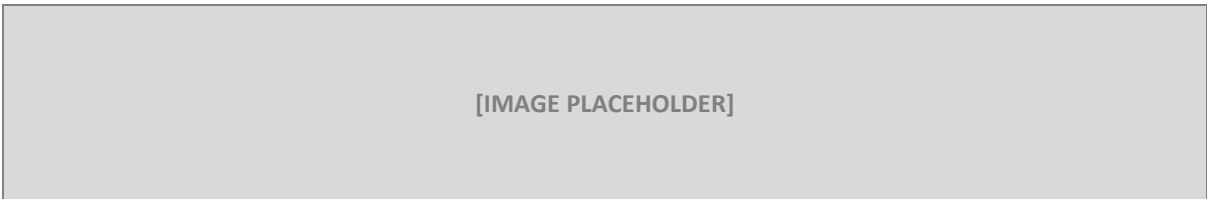


Figure 8-1: [PLACEHOLDER] Full electrical schematic drawing set to be inserted here. Drawing set to include: (1) Main power distribution single-line diagram (415V, 3-phase, 63A incoming); (2) Control circuit schematic - Safety PLC I/O wiring; (3) E-stop dual-channel circuit diagram; (4) Servo drive and STO wiring diagram; (5) KUKA robot controller interface wiring; (6) Maglev conveyor PLC wiring; (7) Cable schedule - all cables tagged with ID, type, routing, and termination points. Source: Indo-MIM electrical design team. Format: A3 drawings at 300 DPI minimum.

8.3 Electrical Specifications Summary

The following key electrical parameters are confirmed from DFM V1.4 and the Machine Safety Assessment Report, pending receipt of full schematics.

Parameter	Specification
Main Supply	415V AC, 3-phase, 50Hz
Maximum Current	63A
Total Connected Power	54 kW (full line)
Station Supply	220V AC per station
Control Voltage	24V DC (Safety PLC and I/O)
Robot Controller	KUKA KR CS Box-2
Safety PLC	FLX3 series
Motion Control	Advantech IPC + PCIE-1203 series motion card
Communication Protocol	EtherCAT (motion), TCP/IP (MES/TRACE), OPC-UA
Main Isolator	Lockable rotary type, 63A, located at main panel
Protection Standard	IEC 60204-1:2016

WARNING - High Voltage Panel [ISO 7010 W003]

The main electrical panel operates at 415V, 3-phase, 63A. Access to this panel is restricted to authorized electricians only. Always complete the full LOTO procedure including LP-01 isolation and voltage verification before opening any electrical enclosure on this machine.

9. PNEUMATIC SCHEMATIC

9.1 Purpose

This section provides the pneumatic circuit documentation for the BG Snap Assembly Machine, covering the compressed air supply, pressure regulation, solenoid valves, and all pneumatic actuators across the six stations.

9.2 Pneumatic Schematic Drawings

OPEN ACTION - OA-P4-010

Pneumatic schematic drawings for the BG Snap Assembly Machine have not been provided at the time of preparation of this manual. Responsibility: Indo-MIM Engineering | Target: Before manual finalization | Status: OPEN

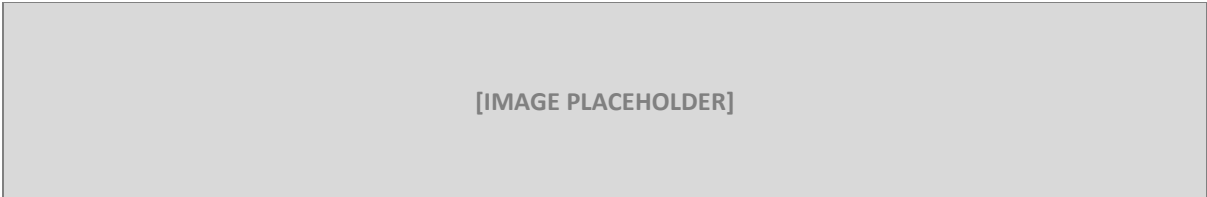


Figure 9-1: [PLACEHOLDER] Full pneumatic circuit diagram to be inserted here. Diagram to include: (1) Main air supply line from site inlet to machine manifold; (2) Pressure regulation and filtration unit (FRL unit) at machine inlet; (3) Solenoid valve bank layout - all stations; (4) Individual actuator circuits - tray locking cylinders (S1, S6), flipping module cylinders (S4), rotating cylinders (S1, S6); (5) Pneumatic symbol legend per ISO 1219-1; (6) Pressure settings table - all regulators and actuators. Source: Indo-MIM pneumatic design team. Format: A3 drawings at 300 DPI minimum.

9.3 Pneumatic Specifications Summary

Parameter	Specification
Supply Pressure	0.5 to 0.7 MPa
Air Quality	Clean, dry, oil-free compressed air
Main Isolation	Manual quarter-turn ball valve with lockout hasp (LP-02)
Actuator Types	SMC and FESTO cylinders - tray locking, flipping, rotating
Solenoid Valves	24V DC solenoid actuation via Safety PLC I/O
Fail-Safe Design	Pneumatic system dumps to safe state on E-stop activation
Design Standard	ISO 4414 - Pneumatic Fluid Power: Safety Requirements

WARNING - Residual Pneumatic Pressure [ISO 7010 W017]

Residual pneumatic pressure remains in downstream lines after LP-02 isolation until the system has fully bled down. Always verify the pressure gauge reads zero before reaching near any pneumatic actuator. Do not assume the system is depressurized simply because LP-02 is closed.

10. HYDRAULIC SCHEMATIC

10.1 Applicability

NOT APPLICABLE

The BG Snap Assembly Machine does not use any hydraulic systems. All motion functions are achieved through servo-driven electric axes, KUKA robots, and pneumatic actuators. No hydraulic circuits, hydraulic power units, or hydraulic actuators are present on this machine.

This section is retained in the manual structure for completeness of the standard documentation framework.

11. HMI ALARMS & FAULT CODES

11.1 Purpose

This section documents the HMI alarm codes generated by the BG Snap Assembly Machine control system, together with the probable cause and recommended operator response for each alarm condition.

The machine is equipped with five HMI stations - one per major section of the line. All alarms are displayed on the relevant station HMI and simultaneously logged in the machine's SQL-based production database for traceability and maintenance trend analysis.

11.2 Alarm Priority Levels

The machine uses three alarm priority levels as defined below.

Priority Level	Colour	Description	Machine Response
Level 1 - Critical	Red	Safety-related or machine-stopping fault. Immediate attention required.	Machine stops automatically. Cycle cannot restart until alarm is resolved and acknowledged.
Level 2 - Warning	Amber	Non-critical condition requiring attention within the current shift.	Machine continues operating. Operator must acknowledge and monitor.
Level 3 - Information	Blue	Operational notification - no action required immediately.	Machine continues operating. Logged automatically.

11.3 HMI Alarm Code List

OPEN ACTION - OA-P4-003

The complete HMI alarm code list for the BG Snap Assembly Machine has not been provided at the time of preparation of this manual. Responsibility: Indo-MIM Engineering | Target: Before manual finalization | Status: OPEN

[IMAGE PLACEHOLDER]

Figure 11-1: [PLACEHOLDER] Full HMI alarm code table to be inserted here. Table to include: Alarm Code | Alarm Description | Station | Priority Level (1/2/3) | Probable Cause | Operator Action | Reset Procedure. Source: Indo-MIM software engineering team - to be extracted from machine C# application alarm module and SQL alarm table.

11.4 General Alarm Response Procedure

Until the complete alarm code list is available, operators must follow this general procedure when any alarm is displayed on the HMI.

Step 1 - Read the Alarm

Note the alarm code, description, and station displayed on the HMI. Do not acknowledge or clear the alarm before understanding what it means.

Step 2 - Assess the Priority

Identify the alarm priority level from the colour indicator. For Level 1 Critical alarms, stop all manual interaction with the machine immediately and contact the maintenance technician.

Step 3 - Do Not Override

Never use Manual Mode or maintenance bypass to override an active alarm without first identifying the cause. Bypassing alarms without understanding the root cause may result in machine damage or personal injury.

Step 4 - Contact Maintenance

For any alarm not covered in the operator's trained knowledge, contact the maintenance technician immediately. Do not attempt to diagnose or repair electrical or mechanical faults without authorization.

Step 5 - Record the Alarm

Record the alarm code, time, and any observations in the machine logbook regardless of whether the alarm was resolved. This record supports maintenance trend analysis.

WARNING - Cleared Alarm is Not a Resolved Alarm [ISO 7010 W019]

A cleared alarm is not a resolved alarm. Acknowledging an alarm on the HMI removes it from the display - it does not fix the underlying condition. If the same alarm recurs within the same shift, do not clear and continue. Stop the machine and contact the maintenance technician.

12. SAFETY FEATURES & INTERLOCKS

12.1 Purpose and Scope

This section tells you what is protecting you, how it works, and what happens when it activates. Read it. The safety system on this machine is well-engineered - but it only works if the people operating and maintaining it understand it.

This section must be read by all maintenance technicians, safety personnel, and authorized service engineers before performing any work on or near the machine's safety systems. Operators are not permitted to modify, bypass, or interfere with any safety device or interlock under any circumstances.

This section has been prepared with reference to the Machine Safety Assessment Report (V1.1, dated 23-March-2026, prepared by Independent Safety Consultant) and DFM V1.4.

12.2 Safety Architecture Overview

The BG Snap Assembly Machine uses a layered safety architecture consisting of four levels of protection working together to reduce risk to an acceptable residual level.

Level 1 - Inherently Safe Design

Machine layout, enclosed station housings, and Maglev conveyor routing minimize the need for operator access to hazardous zones during normal operation. The single operator interaction point is limited to tray loading and unloading at defined access zones with physical boundary separation.

Level 2 - Fixed Guarding

All stations are enclosed within fixed metal guarding panels. Fixed guards are fastened with tools and cannot be removed without deliberate action. All fixed guards must be reinstalled and fully fastened after any maintenance activity before the machine is returned to service.

Level 3 - Interlocked Safeguarding

All access doors are interlocked with the Safety PLC. Opening any interlocked door during machine operation triggers a safety-rated stop. Safety light curtains at the tray loading and unloading zones provide area sensing without physical barriers at the operator interaction points.

Level 4 - Complementary Protective Measures

Emergency stop pushbuttons at all six stations provide a last-resort stop function. Warning labels, PPE requirements, operator training, and this manual form the final layer of risk reduction for residual risks that cannot be fully eliminated by the first three levels.

12.3 Safety Control System

The machine's safety functions are managed by a dedicated Safety PLC operating independently from the machine control IPC.

Parameter	Details
Safety PLC	FLX3 series
Safety Standard	ISO 13849-1:2015
Architecture	Dual-channel, self-monitoring
Safety Outputs	Dual contactor control - servo drives and Maglev conveyor
STO Function	Safe Torque Off on all HIWIN and Toyo servo drives
Robot Safety Interface	KUKA robot safety I/O - ISO 10218-1 compliant
E-Stop Circuit	Dual-channel hardwired, monitored by Safety PLC
Safety Door Circuit	Dual-channel per door, monitored by Safety PLC
Light Curtain Interface	Direct Safety PLC input - OSSD dual-channel
Diagnostic Coverage	High DC per IEC 62061

The Safety PLC operates independently from the machine software application. A fault or crash in the machine's C# software application does not affect the Safety PLC's ability to execute safety functions. The Safety PLC monitors all safety inputs continuously and acts on any fault condition regardless of the state of the machine application.

12.4 Safety Devices - Full Inventory

OPEN ACTION - OA-P4-013

The safety device inventory below is derived from DFM V1.4 and the Machine Safety Assessment Report. Physical installation, cable routing, device mounting positions, and label verification must be confirmed during dry run at Tata Electronics, Hosur after electrical and pneumatic drawing review. This table to be updated and finalized post dry run before manual V2.0 is released. Responsibility: Ravikumar, Certified Safety Professional - site visit | Target: Dry run | Status: OPEN

Device ID	Device Type	Brand & Model	Station	Perf. Level	Category	Function
SD-01	Safety door latch	Keyence GS-51P5	S1	PLe	Cat 4	Interlocks S1 access door - stops machine on opening
SD-02	Safety door latch	Keyence GS-51P5	S2	PLe	Cat 4	Interlocks S2 access door - stops machine on opening
SD-03	Safety door latch	Keyence GS-51P5	S3	PLe	Cat 4	Interlocks S3 access door - stops machine on opening
SD-04	Safety door latch	Keyence GS-51P5	S4	PLe	Cat 4	Interlocks S4 access door - stops machine on opening

Device ID	Device Type	Brand & Model	Station	Perf. Level	Category	Function
SD-05	Safety door latch	Keyence GS-51P5	S5	PLe	Cat 4	Interlocks S5 access door - stops machine on opening
SD-06	Safety door latch	Keyence GS-51P5	S6	PLe	Cat 4	Interlocks S6 access door - stops machine on opening
SD-07	Safety light curtain	Keyence GL-VM39FP	S1	PLe	Cat 4	Area sensing at tray loading zone - stops machine on beam break
SD-08	Safety light curtain	Keyence GL-VM39FP	S6	PLe	Cat 4	Area sensing at tray unloading zone - stops machine on beam break
SD-09	E-stop pushbutton	Per IEC 60947-5-5	S1	PLd	Cat 3	Emergency stop - ES-01
SD-10	E-stop pushbutton	Per IEC 60947-5-5	S2	PLd	Cat 3	Emergency stop - ES-02
SD-11	E-stop pushbutton	Per IEC 60947-5-5	S3	PLd	Cat 3	Emergency stop - ES-03
SD-12	E-stop pushbutton	Per IEC 60947-5-5	S4	PLd	Cat 3	Emergency stop - ES-04
SD-13	E-stop pushbutton	Per IEC 60947-5-5	S5	PLd	Cat 3	Emergency stop - ES-05
SD-14	E-stop pushbutton	Per IEC 60947-5-5	S6	PLd	Cat 3	Emergency stop - ES-06
SD-15	Servo STO	Integrated in servo drives	All	PLd	Cat 3	Safe Torque Off - all HIWIN and Toyo axes
SD-16	Robot safety I/O	KUKA KR CS Box-2	S1, S6	PLd	Cat 3	Robot safety stop - ISO 10218-1
SD-17	Dual contactor	Integrated in panel	All	PLe	Cat 4	Power removal from servo drives and Maglev on safety demand

12.5 Safety Door Interlock Function

The machine is equipped with six interlocked access doors, one per station. Each door interlock uses a Keyence GS-51P5 safety door latch rated at PLe / Category 4 / SIL3 per ISO 13849-1:2015.

Normal Operation - Door Closed:

The door latch is engaged. The Safety PLC receives a healthy dual-channel signal from the door latch. The machine is permitted to run in automatic mode.

Door Opened During Operation:

The door latch dual-channel signal changes state. The Safety PLC detects the change within its monitoring cycle and immediately initiates a safety-rated stop. All servo axes within the affected station decelerate to standstill and STO is applied. The KUKA robot executes a safety stop if the affected station is S1 or S6. The HMI displays a door interlock alarm with the station identifier.

Restart After Door Closure:

The machine does not restart automatically when the door is closed. The operator must physically close and latch the door, confirm on the HMI that the door interlock signal is healthy, acknowledge the alarm, and manually initiate a restart sequence from the HMI. This prevents unexpected restart as required by ISO 14118:2017.

Power-to-Lock Feature:

The Keyence GS-51P5 latch uses a power-to-unlock design. In the event of a power failure or Safety PLC fault, the door latch defaults to the locked state - the door cannot be opened from outside the guarded zone.

WARNING - Emergency Mechanical Door Release [ISO 7010 W003]

In a power failure the door latch defaults to locked. If you are inside the station when power fails, the door will not open from outside. Every station has an emergency mechanical release on the inside of the door. Know where it is before you enter. This is not optional knowledge.

OPEN ACTION - OA-P4-011

Location and operation of emergency mechanical door release on each station to be confirmed and photographed during dry run at Tata Electronics, Hosur. Training on emergency release to be included in maintenance technician induction. Responsibility: Ravikumar, Certified Safety Professional - site visit | Target: Dry run | Status: OPEN

12.6 Safety Light Curtain Function

Two safety light curtains are installed at the tray loading zone (S1) and tray unloading zone (S6).

Parameter	Specification
Device	Keyence GL-VM39FP
Performance Level	PLe
Category	Cat 4
Safety Integrity Level	SIL3
Standard	ISO 13849-1:2015 / IEC 61496-1:2020
Positioning Standard	ISO 13855:2024
Output Signal	OSSD dual-channel to Safety PLC

Parameter	Specification
Indicator	Green = active and unobstructed / Red = beam broken or fault

Normal Operation:

The light curtain transmitter and receiver are aligned and the beam is unobstructed. The Safety PLC receives a healthy OSSD dual-channel signal. The machine is permitted to run with the curtain zone active.

Beam Interrupted During Operation:

When the operator's hand, arm, or any object breaks the light curtain beam during machine operation, the OSSD signal changes state. The Safety PLC detects the change and immediately initiates a safety-rated stop of all hazardous motion within the affected station. The HMI displays a light curtain interrupt alarm.

Restart After Beam Cleared:

The machine does not restart automatically when the beam is cleared. The operator must step clear of the light curtain zone, confirm the green curtain indicator is active, and manually initiate a restart from the HMI.

OPEN ACTION - OA-P4-012

Confirmation of whether light curtain muting is implemented at S1 and S6 is pending site visit verification. If muting is confirmed, muting sensor locations and muting window parameters to be documented and added to this section before manual is finalized. Responsibility: Indo-MIM Engineering + site visit | Target: Dry run | Status: OPEN

12.7 Safe Torque Off (STO) Function

All servo drives on the BG Snap Assembly Machine - including all HIWIN and Toyo series axes across all stations - are equipped with Safe Torque Off (STO).

STO removes the electrical power that generates torque from the motor without removing power from the drive electronics. This means the drive can monitor its own status and communicate with the Safety PLC even when STO is active.

STO is activated automatically in the following conditions:

- Any E-stop pushbutton is activated (ES-01 to ES-06)
- Any interlocked safety door is opened during operation
- Either safety light curtain beam is interrupted during operation
- The Safety PLC detects a safety system fault
- Power is removed from the machine

WARNING - STO Does Not Isolate Electrical Power [ISO 7010 W003]

STO stops the motor. It does not cut power to the drive cabinet. Touch a drive terminal with STO active and you will receive a shock from the DC bus. STO is a machine safety function - not an electrical isolation. Use LOTO for electrical work. Always.

12.8 Robot Safety Function

Both KUKA KR13 R850 robots at S1 and S6 are integrated into the machine safety system per ISO 10218-1:2011.

Safety Function	Implementation
Safety-rated stop	Robot executes a Category 1 controlled stop on safety demand from Safety PLC
Safe speed monitoring	Robot speed is monitored in Manual Mode - maximum 250mm/s teach speed enforced
Safety I/O interface	Hardwired safety I/O between KUKA KR CS Box-2 controller and machine Safety PLC
Working envelope monitoring	Robot working envelope is defined in the KUKA controller - robot cannot move outside defined zone
Teach pendant enable	Three-position enabling device on teach pendant - robot motion in Manual Mode only possible with enabling device held in middle position

DANGER - KUKA Robot Working Envelope [ISO 7010 W028]

The KUKA KR13 moves at up to 2700 degrees per second with a 13 kg payload. In the time it takes you to react, it has already completed its motion. There is no safe distance inside the working envelope while the robot is powered. Complete LP-04 isolation - both S1 and S6 controllers - before entering. Every time.

12.9 Interlock Logic Summary

Safety Condition	Safety PLC Response	Machine Response	Restart Method
E-stop activated (any station)	STO command to all drives	Category 1 controlled stop - all axes	Manual reset + HMI acknowledge
Safety door opened (any station)	STO command to affected station	Category 1 stop - affected station axes	Door closed + HMI acknowledge + manual restart
Light curtain beam broken (S1/S6)	STO command to S1 or S6	Category 1 stop - affected station	Beam cleared + HMI acknowledge + manual restart
Safety PLC fault detected	STO command to all drives	Full machine stop	Safety fault resolved + authorized reset
Power failure	STO active by default	All drives de-energized	Full startup sequence required
Robot safety I/O fault	Robot safety stop	Robot stops - other stations may continue	Robot fault resolved + HMI acknowledge

12.10 Safety Function Testing Requirements

Safety Function	Test Frequency	Test Method	Acceptance Criteria
Safety door interlock - all doors	Weekly	Open each door during a test cycle; verify stop and HMI alarm	Machine stops; alarm displayed; does not restart automatically
Safety light curtain - S1 and S6	Weekly	Break beam with test object during test cycle; verify stop	Machine stops; curtain alarm displayed; green indicator returns on clear
E-stop function - all buttons	Weekly	Activate each E-stop individually; verify controlled stop	All axes stop; STO confirmed; HMI alarm displayed
Dual-channel door interlock integrity	Monthly	Disconnect one channel at Safety PLC; verify fault alarm	Safety PLC fault alarm within 500ms
Dual-channel light curtain integrity	Monthly	Disconnect one OSSD channel at Safety PLC; verify fault alarm	Safety PLC fault alarm within 500ms
STO function on all servo drives	Monthly	Apply STO via Safety PLC; attempt to energize motor; verify no torque	Motor produces no torque with STO active
Robot safety stop function	Monthly	Trigger safety stop via Safety PLC I/O; verify robot controlled stop	Robot stops within defined deceleration time; no fault on restart
Full safety function audit	Every 6 months	Complete ISO 13849-1 verification checklist across all safety functions	All safety functions verified as meeting stated PL and Category

12.11 Prohibited Actions - Safety Systems

The following actions are strictly prohibited. Any of the following will result in the machine being stopped immediately and not returned to service until a full safety review is completed by a Certified Safety Professional or Third Party.

- Bypassing, defeating, or disabling any safety door interlock, light curtain, E-stop, or STO function for any reason
- Entering any guarded zone without completing the full LOTO procedure in Section 4
- Modifying Safety PLC parameters or safety circuit wiring without written authorization from Indo-MIM engineering and the machine manufacturer
- Removing or covering safety warning labels on the machine
- Operating the machine with any fixed guard removed or any access door open
- Using tape, cable ties, or any mechanical device to hold a safety door latch in the closed position while the door is open
- Allowing untrained or unauthorized personnel to operate the machine in any mode

NOTICE - Safety Bypass Reporting

Any deliberate bypass or defeat of a safety function must be reported immediately to the Tata Electronics shift supervisor and Indo-MIM safety team. Any machine found operating with defeated safety functions

will be stopped immediately and will not be returned to service until a full safety review has been completed by a Certified Safety Professional or Third Party.

13. COMPLIANCE & CERTIFICATIONS

13.1 Purpose and Scope

This section documents the regulatory and standards compliance framework applicable to the BG Snap Assembly Machine installed at Tata Electronics, MG Road, Hosur, Tamil Nadu. It identifies all applicable standards, confirms the compliance basis for each major system of the machine, and provides the audit trail required for regulatory inspection and customer compliance verification.

This section is intended for use by safety auditors, regulatory inspectors, Indo-MIM quality personnel, and Tata Electronics EHS representatives. It should be read in conjunction with the Machine Safety Assessment Report (V1.1, dated 23-March-2026, prepared by Independent Safety Consultant) which is held separately by Indo-MIM Limited and Tata Electronics.

13.2 Applicable Standards & Regulations

13.2.1 International Safety Standards

Standard	Title	Relevance to this Machine
ISO 12100:2010	Safety of Machinery - General Principles for Design - Risk Assessment and Risk Reduction	Overall risk assessment methodology - all sections of this manual
ISO 13849-1:2015	Safety of Machinery - Safety-Related Parts of Control Systems	Safety PLC architecture, STO, door interlocks, light curtains, E-stop circuit
ISO 13850:2015	Safety of Machinery - Emergency Stop Function - Principles for Design	E-stop device specification, stop category, reset procedure - Section 5
ISO 14119:2013	Safety of Machinery - Interlocking Devices Associated with Guards	Safety door latch specification and interlock logic - Section 12
ISO 14120:2015	Safety of Machinery - Guards - General Requirements	Fixed guard design and fastening requirements - Section 12
ISO 14118:2017	Safety of Machinery - Prevention of Unexpected Start-Up	Restart prevention logic after E-stop, door open, and light curtain events - Section 12
ISO 13855:2024	Safety of Machinery - Positioning of Safeguards with Respect to Approach Speeds	Safety light curtain positioning at S1 and S6 - Section 12
ISO 10218-1:2011	Robots and Robotic Devices - Safety Requirements for Industrial Robots	KUKA robot safety integration - Section 12
ISO 20607:2019	Safety of Machinery - Instruction Handbook - General Drafting Principles	Structure, content, and language of this operating and safety manual
IEC 82079-1:2019	Preparation of Instructions for Use - Structuring, Content and Presentation	Manual preparation, signal word hierarchy, and usability requirements
IEC 60204-1:2016	Safety of Machinery - Electrical Equipment of Machines - General Requirements	Electrical panel design, wiring, earthing, and protection devices - Section 8

Standard	Title	Relevance to this Machine
IEC 61496-1:2020	Safety of Machinery - Electro-Sensitive Protective Equipment	Safety light curtain specification - Section 12

13.2.2 Indian Standards & Regulations

Standard / Regulation	Title	Relevance
IS 4571:2008	Safety of Machinery (Indian Standard)	Indian regulatory compliance - all sections
IS 12100:2012	Safety of Machinery - General Principles for Design (Indian adoption of ISO 12100)	Risk assessment methodology under Indian standards framework
Factories Act 1948	Indian Industrial Labour Law - Section 21: Fencing of Machinery	Legal basis for machine guarding requirements at Tata Electronics facility

13.2.3 Component-Level Standards

Component	Applicable Standard	Compliance Basis
Safety door latch (Keyence GS-51P5)	ISO 14119, IEC 60947-5-3	Manufacturer's declaration - PLe / Cat 4 / SIL3
Safety light curtain (Keyence GL-VM39FP)	IEC 61496-1, ISO 13849-1	Manufacturer's declaration - PLe / Cat 4 / SIL3
Safety PLC (FLX3 series)	ISO 13849-1	Manufacturer's declaration
KUKA robot (KR13 R850 Z200-2 CS)	ISO 10218-1, IEC 60204-1	Manufacturer's declaration - CE marked
Atlas smart screwdriver (QMC-21-10-HM4)	IEC 60745, ISO 11148 series	Manufacturer's declaration
Deprag screw feeder	IEC 60204-1	Manufacturer's declaration

13.3 Compliance Statement

This manual was prepared by Ravikumar, Certified Safety Professional, based on the Machine Safety Assessment Report (V1.1, March 2026), DFM V1.4 (March 2026), and the standards listed in Section 13.2. Where the machine was not yet available for physical inspection, gaps have been documented as Open Actions - not assumed, not ignored, not filled with generic content. The Open Actions Register is a live document. It closes when the machine is verified, not before.

The manual documents the machine's safety features, operating procedures, and maintenance requirements based on the following sources:

- Machine Safety Assessment Report, V1.1, dated 23-March-2026, prepared by Independent Safety Consultant
- Design for Manufacturing Document (DFM V1.4), dated 18-March-2026, prepared by Indo-MIM Limited

- Site visit and physical verification at Tata Electronics, Hosur - scheduled pending machine readiness
- Operator and maintenance personnel interviews - to be conducted at dry run

Where original technical data was unavailable at the time of manual preparation, professional judgment and industry best practices have been applied. All such areas are formally documented as Open Actions in the working Open Actions Register.

Customer Responsibility: Final review and verification of all information in this manual is the responsibility of Indo-MIM Limited as machine manufacturer and Tata Electronics as end user. All items listed in the Open Actions Register must be reviewed, verified, and signed off by the respective responsible parties before this manual is formally released as V2.0 and the machine is placed into production service.

13.4 Liability Disclaimer

This manual documents what was known at the time of writing. It does not cover modifications made to the machine after this date, operation outside the intended use defined in Section 2.3, or failure to complete the Open Actions in the working register. Ravikumar has documented everything that could be documented with the information available. The rest is the responsibility of Indo-MIM Limited and Tata Electronics.

Specifically, neither the manual author nor the consulting firm assumes liability for injuries or damages resulting from:

- Non-compliance with procedures outlined in this manual
- Failure to maintain the machine according to the maintenance schedule in Section 7
- Operation by personnel not trained per the requirements stated in this manual
- Modification of the machine, its safety systems, or its operating parameters without authorization
- Failure to close Open Actions before placing the machine into service

13.5 Manual Control & Revision History

Revision	Date	Description	Prepared By	Reviewed By
V1.0	April 2026	Initial issue - based on DFM V1.4 and Machine Safety Assessment Report V1.1. Placeholders retained for items pending site visit and dry run.	Ravikumar, Certified Safety Professional	Indo-MIM / Tata Electronics
V1.1	TBD	To be issued following dry run - LOTO verification, safety device locations, lubrication schedule, troubleshooting guide, HMI alarm codes, electrical and pneumatic schematics inserted.	Ravikumar, Certified Safety Professional	Indo-MIM / Tata Electronics
V2.0	TBD	Full release - all Open Actions closed, all placeholders replaced, customer sign-off complete.	Ravikumar, Certified Safety Professional	Indo-MIM / Tata Electronics

13.6 Document Control

Field	Details
Document Title	Operating & Safety Manual - BG Snap Assembly Machine
Document Number	To be assigned by Indo-MIM document control
Revision	V1.0
Classification	Confidential - Indo-MIM Limited / Tata Electronics
Language	English
Format	Word (.docx) - editable / PDF (.pdf) - controlled copy
Storage Location	Indo-MIM document management system + Tata Electronics EHS records
Retention Period	Minimum 10 years from date of machine commissioning per Factories Act 1948
Next Review Date	Following dry run - target V1.1 release

13.7 Customer Sign-Off

The following sign-off is required before this manual is formally released as a controlled document and the machine is placed into production service.

Indo-MIM Limited - Manufacturer Authorization

I confirm that this Operating and Safety Manual has been reviewed by Indo-MIM Limited's engineering and quality team, that all technical content relating to machine design and specifications has been verified as accurate, and that all Open Actions assigned to Indo-MIM have been reviewed and accepted for closure as indicated in the working Open Actions Register.

Name: _____ Designation: _____

Signature: _____ Date: _____

Tata Electronics - End User Authorization

I confirm that this Operating and Safety Manual has been reviewed by Tata Electronics' EHS and maintenance team, that the operating and maintenance procedures are appropriate for the Tata Electronics production environment, and that all Open Actions assigned to Tata Electronics have been reviewed and accepted for closure as indicated in the working Open Actions Register.

Name: _____ Designation: _____

Signature: _____ Date: _____

Certified Safety Professional - Manual Author

I confirm that this Operating and Safety Manual has been prepared in accordance with ISO 12100:2010, ISO 20607:2019, IEC 82079-1:2019, and IS 4571:2008, that all identified hazards from the Machine Safety Assessment Report have been addressed, and that all areas of uncertainty have been formally documented as Open Actions.

Name: Ravikumar Designation: _____

Signature: _____ Date: _____

APPENDIX A — RESIDUAL RISK SUMMARY

A.1 Purpose

The safeguards on this machine are good. They are not perfect. This appendix tells you what risk remains after everything that could be engineered has been engineered. These risks are real. They are low - but only because the people working with this machine respect the procedures. The moment those procedures become shortcuts, the risk rating changes.

All operators, setup technicians, and maintenance technicians must read this appendix in full before working with or near the BG Snap Assembly Machine.

This residual risk summary has been derived from the Machine Safety Assessment Report (V1.1, dated 23-March-2026, prepared by Independent Safety Consultant). The complete risk assessment document is held by Indo-MIM Limited and Tata Electronics and is available for audit upon request.

A.2 Residual Risk Summary Table

Risk Ref	Hazard Description	Residual Risk Level	Persons at Risk	Conditions Under Which Risk Exists	Required Precautions
RR-01	Crushing injury from vertical servo-driven tray lifter mechanism	Low	Maintenance technician	During troubleshooting or maintenance with guarding removed	Full LOTO before access. Never place hands near tray lifter without LOTO complete. Apply mechanical support or locking device to vertical axis before working beneath it.
RR-02	Shearing injury from multi-axis gantry servo system	Low	Maintenance technician	During maintenance access to servo axis zone	Full LOTO before access. Verify STO active on all affected drives before entering servo zone. Maintain minimum safe distance from axis travel path at all times.
RR-03	Entanglement or pinch from fixture assembly moving components	Low	Maintenance technician, Setup technician	During troubleshooting or fixture changeover	Full LOTO before access. Never reach into fixture channel while machine is energized. Use appropriate PPE including cut-resistant gloves where permitted.
RR-04	Impact or trapping from KUKA robot arm or SCARA transfer mechanism	Low	Maintenance technician, Setup technician	During Manual Mode robot teaching or maintenance inside robot envelope	RR-04 is the highest consequence risk on this machine. A robot arm moving at speed inside a confined station is fatal. There is no partial mitigation here - LP-04 isolation is the only acceptable control. If robot

Risk Ref	Hazard Description	Residual Risk Level	Persons at Risk	Conditions Under Which Risk Exists	Required Precautions
					teaching requires the robot to be powered, a second authorized person must be present at the controller at all times, with their hand on the E-stop. Never enter the robot envelope alone with the controller powered.
RR-05	Pinch injury from tray locking pneumatic cylinder unexpected actuation	Low	Maintenance technician	During troubleshooting of pneumatic system with machine partially energized	Full LOTO including pneumatic isolation (LP-02) and pressure bleed-down to zero before accessing pneumatic actuators. Verify zero pressure on gauge before reaching near cylinder.
RR-06	Trapping or shearing from linear guideways and moving carriages	Low	Maintenance technician	During axis maintenance with servo drives accessible	Full LOTO before access. Verify STO active. Do not place fingers or tools between carriage and rail during any axis movement.
RR-07	Entanglement from winder assembly rotating components	Low	Maintenance technician, Setup technician	During setup or troubleshooting of tape reel feeder mechanism	Full LOTO before accessing winder assembly. Never attempt to clear a tape jam with the machine energized. Use appropriate PPE.
RR-08	Pinch or trapping from robot EOAT gripper mechanism	Low	Maintenance technician	During robot EOAT inspection or replacement	Full LOTO including robot controller isolation (LP-04). Never actuate gripper manually without confirming pneumatic pressure is zero.
RR-09	Crushing from gravity-assisted downward movement of vertical lifting actuators	Low	Maintenance technician	During troubleshooting of lifting actuator with machine partially energized	Apply mechanical support or locking pin to vertical axis before working beneath it. Full LOTO required. Do not rely on servo brake alone to hold vertical load during maintenance.
RR-10	Impact or pinch from moving trays and fixtures during manual handling	Low	Maintenance technician, Operator	During manual tray loading, unloading, or fixture changeover	Follow defined tray loading procedure in Section 6.7. Never reach beyond the tray loading slot boundary. Wear appropriate PPE for manual handling tasks.

Risk Ref	Hazard Description	Residual Risk Level	Persons at Risk	Conditions Under Which Risk Exists	Required Precautions
RR-11	Electric shock from live parts inside electrical enclosure during maintenance	Low	Maintenance technician, Authorized electrician	During electrical maintenance with panel open	Full LOTO including main electrical isolation (LP-01). Verify voltage absent using calibrated tester before touching any electrical component. Wear insulated gloves and face shield for all electrical work. Authorized electrician only.
RR-12	Electric shock from residual stored energy in servo drive capacitors after power isolation	Low	Maintenance technician, Authorized electrician	During electrical maintenance immediately after power isolation	Wait minimum 60 seconds after LP-01 isolation before touching servo drive components. Verify DC bus voltage has discharged using calibrated meter before contact.
RR-13	Awkward posture and physical strain during manual tray loading and unloading	Low	Operator	During every production shift - tray loading at S1 and unloading at S6	Tray weight to be confirmed - see OA-P4-004. Until confirmed, treat every loaded tray as a two-person lift if it feels heavy. Musculoskeletal injuries from repetitive tray handling are cumulative - they do not announce themselves until the damage is done. Report any discomfort to your supervisor early.
RR-14	Unexpected machine start-up after power restoration or fault reset	Low	All personnel	During or after power restoration, fault reset, or control system restart	Always confirm machine is in safe state and all personnel are clear before acknowledging any restart prompt on HMI. Never acknowledge a restart without performing a full visual check of the machine perimeter.
RR-15	Hazardous motion occurring if safety interlock is deliberately bypassed	Low	All personnel	If any safety door interlock, light curtain, or STO function is bypassed or defeated	Safety bypasses are strictly prohibited - refer to Section 12.11. Any bypass must be reported immediately to shift supervisor and EHS team. Machine must not be returned to service until bypass is removed and safety function is verified.

A.3 Personal Protective Equipment (PPE) Requirements

The following PPE must be available at the machine at all times and must be used by all personnel performing the tasks indicated.

PPE Item	Required For	Standard
Safety shoes (steel toecap)	All personnel entering the production area	IS 15298 / EN ISO 20345
Cut-resistant gloves	Maintenance technicians accessing fixture components, snap feeders, and tape reel mechanisms	EN 388 Level 3 minimum
Insulated electrical gloves	Authorized electricians during electrical panel maintenance	IEC 60903 Class 00 minimum
Safety glasses	All personnel during maintenance activities involving compressed air cleaning	IS 5983 / EN 166
Face shield	Authorized electricians during electrical panel work	EN 166
High-visibility vest	Maintenance technicians working in production area during shift	IS 15809

NOTICE - PPE is the Last Line of Defence

PPE is the last line of defence. It supplements but does not replace engineering safeguards, LOTO procedures, and safe working practices. Wearing PPE does not make it safe to bypass any safety function or deviate from the procedures in this manual.

APPENDIX B — MAINTENANCE LOG TEMPLATE

All completed maintenance tasks must be recorded below. One entry per task. File completed log sheets in the machine maintenance logbook kept at the machine.

Date	Task #	Task Description	Technician Name	Result	Observations	Signature

Result codes: S = Satisfactory | R = Item replaced | F = Fault found | D = Deferred

NOTICE - Legal Retention Requirement

Maintenance records are a legal document under the Factories Act 1948. Retain all completed log sheets for a minimum of three years. Records must be available for inspection by statutory authorities, Indo-MIM quality team, or a Certified Safety Professional or Third Party upon request.

APPENDIX C — LOTO LOG TEMPLATE

A new entry must be made for every LOTO event without exception. The completed log must remain at the machine at all times. Do not remove log sheets until the retention period has elapsed.

Date	Time Out	Technician Name	Stations / Points Locked Out	Reason for LOTO	Time In	All Padlocks Removed	Technician Signature

Stations / Points column: List all isolation points applied - e.g. LP-01, LP-02, LP-04 S1, LP-05

DANGER - Lost Padlock Key Protocol [ISO 7010 P010]

If a padlock cannot be removed at Tag-In because the keyholder is unavailable, do not cut the lock without written authorization from the Indo-MIM maintenance supervisor and Tata Electronics EHS representative. Document the exception in the Observations column and attach a written explanation to this log sheet.